Long Read RNA Sequencing in Primary Lung Cell Types Reveals Principles of Nonsense-Mediated Decay

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Introduction

Nonsense-mediated decay (NMD) is a highly conserved mechanism in which RNA transcripts are targeted for degradation after ribosomal engagement. One of the functions of NMD is as a "proofreader" that identifies and degrades transcripts that contain premature termination codons (PTCs). However, NMD can be triggered for reasons beyond PTCs, including long 3' UTRs[1] and the presence of upstream open reading frames[2]. In general, NMD-triggering events lead to stalled or delayed translation, and while some of the RNA sequence characteristics that cause NMD are known, many remain unknown. Consequently, our ability to predict the extent to which specific transcripts will be degraded by NMD remains imperfect, despite the widespread use of heuristics such as the 50-nucleotide rule. The ability to predict NMD susceptibility of transcripts is important, because it has been shown that NMD is broadly active and degrades ~15% of the RNA output of protein coding genes[3].

In addition to its role as transcript proofreader, NMD has been integrated into other biological processes as a means of rapid regulation of protein levels. For example, NMD plays a major role in cellular stress[4], where it provides baseline suppression of stress response genes. When cells experience stress, NMD is inhibited, resulting in a rapid increase in stress response RNAs, providing clear evidence that cells have utilized NMD as a regulatory mechanism. Because NMD can be triggered by non-PTC-based sequence features, a subset of NMD-targeted transcripts, such as the master regulator of the integrated stress response - *ATF4*, are capable of producing intact, functional protein[5].

We hypothesized that long read RNA sequencing coupled with NMD inhibition would reveal specific NMD-targeted isoforms in lung cell types and that RNA sequence based models trained on these data would reveal principles of isoform-level susceptibility to NMD. To study this hypothesis, we performed transcriptome-wide discovery of NMD susceptible transcripts in four different primary lung cell types – large airway epithelial (LAE), microvascular endothelial (MV), fibroblast (FB), and alveolar type 2 (AT2) – using short and long-read RNA sequencing in normal culture conditions and in the presence of NMD inhibition. We identified tens of thousands of transcripts that are degraded by NMD, confirming that NMD is widespread and affects highly expressed genes and transcripts. We observed significant enrichment of NMD susceptible genes involved in the integrated stress response and alternative splicing, as well as more cell type-specific pathways and processes. Using deep learning, we trained a highly accurate model to predict NMD susceptibility from transcript sequence, identifying novel RNA sequence features responsible for NMD, including sequence elements at the stop codon that promote readthrough translation.

Methods

Isolation and Culturing of Primary Lung Cell Types:

Human lung tissue not used for transplant was obtained, with informed consent, under protocol 03-1396, approved by the University of North Carolina at Chapel Hill Biomedical Institutional Review Board. LAE cells were obtained as previously described[6,7]. MV and FB cells were also obtained from the same donors as previously described[8]. MV, FB, and AT2 cells were obtained from the same three donors. LAE cells were obtained from four different donors and were cultured at 37C, 5% CO_2. MV cells were obtained by dispase and elastase digestion of peripheral lung tissue with the visceral pleura removed and were cultured on type I/III collagen-coated culture wells in EGM-2 media (Lonza, CC-3162) supplemented with FBS. Cells were enriched through two to three rounds of CD31 bead purification (Dynabeads; Life Technologies/ Invitrogen), achieving >95% CD31 positivity by flow cytometry, and were used between passages 5 and 10. FB cells were derived by outgrowth from finely minced distal lung tissue plated onto scratched type I/III collagen-coated dishes in Dulbecco's modified Eagle medium with high glucose (DMEMH) with 10% FBS, antibiotics, and antimycotics. After initial outgrowth, cells were passaged with trypsin/EDTA and sub-cultured under the same conditions. Fibroblast identity was confirmed by morphology and cells were negative for CD31 and pan-cytokeratin which was assessed via flow cytometry and immunofluorescence staining. Additionally, passage one LAE cells procured as described[6], were cultured in modified SAGM complete medium (CC3119, Lonza), supplemented with 1 μM A8301, 1 μM DMH-1, 1 μM CHIR99021, 10 μM Y-27632, and penicillin/streptomycin. AT2 cells were obtained by tissue mincing, enzymatic digestion, and HT2-280 antibody magnetic bead purification as described previously[9]. Cells were cultured and expanded as alveolospheres in Matrigel pellets to passage number 2-3. Alveolospheres were passaged and cultured as monolayer on 5% Matrigel followed by SMG1i or DMSO treatment.

All cultured cells were incubated with 0.3uM of SMG1i or an equivalent volume of DMSO (control), each in their respective media, and cultured at 37C, 5% CO_2 for six hours. After LAE, MV and FB cell cultures were washed with PBS, cells were lysed in RLT Plus buffer (RNeasy Plus Mini Kit, Qiagen 74134). After SMG1i or DMSO treatment AT2 cells were washed once with PBS and lysed in RLT Plus buffer. All lysates were stored at -80C until shipping on dry ice.

Short-read RNA Sequencing

cDNA libraries were constructed from polyA-selected, paired end stranded libraries using the NEBNext ULTRAEXPRESS® RNA Library Prep Kit and sequenced on a Illumina NovaSeq 6000 at the Maryland Genomics core sequencing facility. The mean RIN value for processed samples was 9.3. Libraries were sequenced to an average depth of 59 million reads. RNA-seq data were processed through the nf-core RNA sequencing pipeline (https://nf-co.re/rnaseq/) version 3.14.0 using Nextflow 24.04.4. Read quality and trimming was performed with Trim Galore 0.6.7 (cutadapt 4.6) with default parameter values. Alignment quality and other QC metrics were generated and evaluated with FastQC 0.12.1, samtools 1.17, and MultiQC. Stranded reads were aligned to the human GRCh38 reference genome

(GRCh38.primary_assembly.genome.fa.gz) using the STAR aligner 2.7.10a in two-pass mode and the GENCODE 49 GTF (gencode.v49.primary_assembly.annotation.gtf) with default parameters. Junctional reads were assigned according to default parameters. For gene and isoform level quantification, isoform counts were estimated with Salmon version 1.10.1 and default parameters. Gene and isoform level counts were generated with the tximport Bioconductor package (bioconductor-tximeta 1.12.0, r-base 4.1.3). The counts used for this analysis were generated with tximport setting the parameter *counts from abundance* = “length scaled TPM”.

Long-read RNA Sequencing

cDNA PacBio Kinnex libraries were constructed from isolated RNA according to PacBio protocol 103-238-700 REV07 with the following modifications: 1.0X SMRTbell cleanup beads were used instead of 0.9X to avoid enrichment of long transcripts in order to achieve properly formed concatenated libraries. Prior to concatenation, each of the eight Kinnex PCR reactions was pooled by equal mass based on their Qubit concentration, instead of including a fixed volume from each tube. The Kinnex FL RNA libraries were sequenced on a PacBio Revio instrument with Revio SMRT cells and SPRQ chemistry to an average depth of 13 million aligned FLNC reads. Long reads were aligned to the human genome (GRCh38.primary_assembly.genome.fa) using minimap2 (minimap2 -ax splice:hq --junc-bed known_junctions.bed --MD --cs=long -uf GRCh38_splice.mmi input.fastq.gz | samtools sort -o out.bam)[10]. Quantification was performed using the PacBio Isocall pipeline (https://github.com/PacificBiosciences/isocall). Long-read isoform assemblies from isocall were classified with SQANTI3 QC v5.5.4[11]. Because TransDecoder2 dominates runtime, we ran SQANTI3 QC in parallel across chromosomes via a custom Nextflow pipeline (https://github.com/peter4244/sqanti3_by_chromosome) that removes malformed-exon transcripts, splits the isocall GTF by chromosome, and invokes sqanti3_qc.py per chromosome using the following command structure: sqanti3_qc.py --isoforms <chrom>.gtf --refGTF gencode.v49.primary_assembly.annotation.gtf --refFasta GRCh38.primary_assembly.genome.fa --CAGE_peak refTSS_v3.3_human_coordinate.hg38.sorted.updated.bed --polyA_motif_list polyA.list.txt --fl_count nmd_isocall.count_matrix.txt --report skip --novel_gene_prefix <chrom> -t <cpus> -o <chrom> -d . SQANTI3 filter and rescue modules were not applied.

Isoform Landscape Characterization

Characterization of the long-read isoform landscape was performed on DMSO samples only (n = 162,800, see filtering description in Differential Gene and Transcript Expression section below). An isoform was considered expressed in a given cell type if it had a summed count ≥1 across that cell type's DMSO samples. Cell type-restricted isoforms were defined as those detected in only one of the four cell types under this criterion. Pairwise isoform-set overlap across cell types was quantified using the Jaccard index on binary detection vectors. Pairwise quantitative similarity was quantified using Spearman and Pearson correlations of mean log2-CPM values (computed with edgeR::cpm using log = TRUE and prior.count = 1) across isoforms for each cell type under DMSO.

To identify isoforms whose baseline expression distinguishes the four cell types, we performed a one-vs-rest differential expression analysis on DMSO samples only. The long-read isoform DGEList and the short-read gene DGEList were each restricted to DMSO samples from AT, LAE, FB, and MV, and a DMSO-restricted filterByExpr was re-applied (min.count = 5, min.total.count = 10) using the cell type design matrix, retaining 105,938 isoforms and 21,547 genes for testing. Library sizes were re-normalized via TMM, and voom transformation was applied. The design formula was ~ 0 + cell_type without a treatment term (DMSO being the only condition). Donor was incorporated as a duplicateCorrelation blocking; the estimated within-donor correlation was 0.026. For each cell type, a one-vs-rest contrast was constructed as that cell type minus the mean of the other three (e.g., LAE − (AT + FB + MV)/3), fit via contrasts.fit, and tested with empirical Bayes moderation via eBayes. Isoforms or genes with BH-adjusted $p < 0.05$ were classified as significantly associated with a given cell type. To borrow strength across the four one-vs-rest contrasts, which are non-independent by construction, we additionally applied multivariate adaptive shrinkage (mashr) to the limma contrast estimates following the same procedure described below for the SMG1i-vs-DMSO analyses. Effect sizes were taken from contrast coefficients and standard errors from coefficient/moderated-t ratios. The null correlation matrix $\hat{v}$ is estimated from a 20,000-feature random subset using estimate_null_correlation_simple. Covariance matrices included both data-driven (PCA-based on strong signals from univariate mash_1by1 at lfsr < 0.05, up to 5 PCs) and canonical structures via cov_canonical. Posterior mean log2 fold changes and local false sign rates were extracted, and features with lfsr < 0.05 and posterior mean log2 fold change > 1 were classified as cell type markers.

<u>Differential Gene and Transcript Expression</u>

To identify genes and isoforms regulated by NMD, we performed differential expression analysis at the gene level using both short-read and long-read data, and at the isoform level using long-read data. The analysis was restricted to four primary lung cell types: AT2, LAE, FB, and MV. For all analyses, raw counts were organized into DGEList objects using the edgeR Bioconductor package. Low expressed features were removed using filterByExpr with a minimum count threshold of 5 and a minimum total count of 10. Library sizes were normalized using the trimmed mean of M-values (TMM) method. After filtering and normalization, the short-read gene-level analysis tested 25,955 genes, and the long-read isoform-level analysis tested 162,800 isoforms. For each analysis, we fit a linear model using limma-voom with the design formula ~ cell_type + treatment + cell_type:treatment, where LAE was used as the reference level for cell type and treatment was SMG1i versus DMSO. To account for the paired donor structure of the experiment (each donor contributing matched SMG1i and DMSO samples), we estimated within-donor correlation using duplicateCorrelation and incorporated this as a blocking factor in the model fit via lmFit. Cell type-specific treatment effects were estimated by extracting contrasts for the SMG1i effect within each cell type using contrasts.fit, followed by empirical Bayes moderation via eBayes.

We then applied multivariate adaptive shrinkage (mashr) [12] to the limma-voom contrast estimates. For each analysis level, effect sizes (log2 fold changes) were taken directly from the contrast coefficient matrix, and standard errors were derived from the ratio of coefficients to moderated t-statistics. Features with non-finite or non-positive standard errors in any cell type were excluded. The null correlation structure across cell types was estimated from a random subset of features using estimate_null_correlation_simple. Covariance matrices were specified using both data-driven (PCA-based, derived from strong signals identified by univariate mash_1by1 analysis at lfsr < 0.05) and canonical covariance structures via cov_canonical. The mashr model was fit using the mash function with both sets of covariance matrices. Posterior mean log2 fold changes and local false sign rates (lfsr) were extracted for downstream analysis. Genes or isoforms with lfsr < 0.05 and posterior mean logFC > 0 were classified as NMD susceptible, indicating transcripts whose expression increases upon NMD inhibition and are therefore subject to degradation by NMD under normal conditions.

Cell Type Specificity

To assess the cell type specificity of NMD, every NMD susceptible feature (lfsr < 0.05, posterior mean logFC > 0) was classified as either shared (NMD susceptible in at least one other cell type at the same threshold) or cell type-specific (NMD susceptible in only that cell type). To quantify the similarity of NMD effect sizes across cell types, we computed pairwise Pearson correlations of mashr posterior mean logFC values among NMD susceptible features.

Gene set enrichment analysis (GSEA) was performed on the short-read gene-level mashr posterior mean logFC rankings using fgsea with MSigDB pathway collections (Hallmark, KEGG, Reactome, GO-BP). For each cell type, genes were ranked by signed posterior mean logFC and tested against pathway gene sets with a minimum size of 15 and maximum size of 500. Pathways with FDR < 0.05 were considered significant.

GSEA was also performed at long-read isoform resolution. Because MSigDB pathway sets are defined at the gene level, each gene was assigned the signed mashr posterior mean log2 fold change of its largest-effect isoform (the isoform NMD susceptible isoform with the largest logFC) and genes were ranked by this isoform-anchored statistic. fgsea was then run against the same MSigDB collections (Hallmark, KEGG, Reactome, GO-BP) with identical gene-set size limits (minimum 15, maximum 500) and the same FDR < 0.05 significance threshold, separately for each cell type.

Reanalysis of Tan et al. (2025)

The recent study by Tan et al [13] also reports sharing of cell type-specific NMD effects across different cell types. To compare these estimates, cross-cell type sharing on a more equivalent footing between their study and ours, we reanalyzed the transcript-level EBSeq results from Tan et al. Supplementary Tables S1, S2, S4, and S6 (UPF2-KD, UPF2-iKO, UPF2-iKO&KD, and UPF3B-KO, each in hESCs and NPCs) with the same mashr framework used here, in place of the independent per-condition thresholding approach used in that study. For each transcript and condition we set the effect size to $\hat{b} = \log_2(\text{PostFC})$ and approximated its standard error as $\hat{s} = |\log_2(\text{PostFC})| / \Phi^{-1}(1 - \text{PPEE}/2)$, treating the EBSeq posterior probability of equivalent expression (PPEE = 1 − PPDE) as a two-sided significance proxy for weighting only; transcripts with PostFC ≤ 0 or = 1, PPEE ≥ 0.99, or non-positive $\hat{s}$ were excluded. Mashr was fit jointly across the eight conditions as described for our primary analyses (data-driven plus canonical covariances; null correlation from a random feature subset), yielding posterior mean $\log_2$ fold changes and local false sign rates. A transcript was called an NMD target in a condition if lfsr < 0.05 and posterior mean > 0; UPF2-dependent targets per cell type were the union of the three UPF2 conditions, and cross–cell type overlap was computed as the shared transcript count relative to each cell type's set (matching Tan et al.). For comparison we also reproduced the original threshold-based overlap using PostFC cutoffs (>2 and >1.5).

<u>Transcriptional Output Lost and Productive Isoform Response</u>

To quantify the overall transcriptional burden of NMD, we developed a measure termed percent transcriptional output lost. For each cell type and donor, the metric was computed as the summed positive expression differences (SMG1i minus DMSO) across all NMD susceptible isoforms divided by the total SMG1i expression where $\Delta_{i,d}^{+} = \max(CPM^{SMG1i}-CPM^{DMSO}, 0)$ and D_C is the set of matched donors for cell type, c. The same measure was computed at the gene level using gene-aggregated CPM values. Per-isoform fraction lost was computed as (SMG1i CPM − DMSO CPM) / SMG1i CPM for each NMD susceptible isoform with $\Delta_{i,d}^{+}$.

$$\%\ \text{Output Lost}_c = 100 \bullet \frac{\sum_{i=1}^{n} \sum_{d \in D_c} \Delta_{i,d}^{+}}{\sum_{i=1}^{n} \sum_{d \in D_c} CPM_{i,d}^{SMG1i}}$$

To measure how the productive (non-NMD) isoforms of a gene respond when its NMD susceptible isoforms accumulate, we restricted to protein-coding genes (≥1 SQANTI3 annotated coding isoform) and, within each cell type, partitioned each gene's isoforms into a productive compartment and an unproductive compartment. Specifically, the unproductive compartment quantification for each gene in each sample was the sum of expression of all of the NMD susceptible isoforms (mashr lfsr < 0.05, posterior mean log2FC > 0). At the gene level, genes were also labeled as NMD genes (containing ≥1 confident unproductive isoform), non-NMD genes (lfsr>0.2), or uncertain (borderline-up at 0.05 ≤ lfsr < 0.20 or lfsr < 0.05 and log2FC < 0).

Because the accumulation of unproductive isoforms under SMG1i artificially deflates the counts for every productive isoform, we applied a custom normalization that removes this artifact. For gene g with productive isoforms prod(g) and unproductive isoforms unprod(g), the adjustment proceeds in three steps. First, the productive compartment under DMSO was summed:

$$P_g^{DMSO} = \sum_{i \in prod(g)} \overline{CPM}_i^{DMSO}$$

Second, each unproductive isoform was assigned to the proportion it occupied within that gene under DMSO for each sample, s, and scaled to the current SMG1i productive compartment of that gene.

$$share_i = \frac{\overline{CPM}_i^{DMSO}}{P_g^{DMSO}}\ ,\ i \in prod(g)$$

$$\overline{CPM}_{i,s} = share_i \sum_{j \in prod(g)} CPM^{SMG1i}_{j,s}$$

Finally, each adjusted library size was rescaled to one million.

$$CPM^{adj}_{i,s} = \overline{CPM}_{i,s} \bullet \frac{10^6}{\sum_k \overline{CPM}_{k,s}}$$

Unproductive isoforms lacking a positive DMSO productive compartment were held at their DMSO mean. Summing CPM_adj over a gene's productive isoforms gives the adjusted productive CPM, prodCPM_SMG1i_adj, in which the contribution of the unproductive surge has been removed. Throughout, prodCPM and unprodCPM denote the per-gene sums of CPM over prod(g) and unprod(g), totalCPM = prodCPM + unprodCPM, and the superscripts DMSO, SMG1i_raw, and SMG1i_adj denote values under DMSO, uncorrected SMG1i, and NMD-anchored-adjusted SMG1i, respectively. We verified that this normalization left control (non-NMD) genes essentially unchanged between DMSO and adjusted SMG1i (Spearman ≥ 0.96, slope ≈ 1) and preserved gene rankings relative to the uncorrected analysis (Spearman ≥ 0.99 between adjusted and raw fold changes).

From these compartments we derived two per-gene, per-cell-type quantities (means across donors), writing prodCPM and totalCPM for the productive and total gene CPM and f = prodCPM/totalCPM for the productive fraction:

$$\text{Adjusted Productive logFC} = log_2\left(\frac{prodCPM^{SMG1i,\,adj} + 1}{prodCPM^{DMSO} + 1}\right)$$

$$\text{Productive-only Output Shift} = \frac{prodCPM^{SMG1i,\,adj} - prodCPM^{DMSO}}{CPM^{DMSO}}$$

Productive direction was tested per cell type with limma-trend (eBayes) on the adjusted productive logFC, with treatment as the predictor and within-donor correlation incorporated via duplicateCorrelation, then jointly shrunk across cell types with mashr (data-driven plus canonical covariances; null correlation from a random feature subset) exactly as for the primary analyses. Genes with lfsr < 0.05 and posterior mean > 0 (< 0) were called productive-up (productive-down).

To determine whether the productive response analysis captured reproducible signal rather than noise, we performed a series of donor- and platform-based tests operating on the donor-paired adjusted productive log2 fold changes. The Rademacher sign-flip permutation test built the "direction is random" null by flipping donor signs 2,000 times and comparing the number of consistent within-donor effects to the permutation null [14], summarized as an empirical false-discovery rate (expected/observed). Cross-donor sign concordance was compared to its exact binomial chance expectation, 2(1/2)^n for n donors (n=3 or 4). The fraction of genes carrying a real effect was estimated as 1 − π0 with Storey's estimator (tuning parameter λ = 0.5) [15].

We used logistic regression to ask what distinguishes the direction of a gene's productive response. Among responsive genes, we modeled the log-odds that a gene's productive fraction moved up rather than down as a function of: NMD status; baseline total expression (log10 CPM); unproductive-isoform expression at DMSO and at SMG1i (log10 CPM); the SMG1i-induced gain in unproductive expression; the number of isoforms per gene (log10); and short-read gene induction. Cell type was included as a nuisance covariate. Separately, to ask whether a gene responded at all in a given direction, we fit one-versus-rest logistic regressions on the same predictors. The predictive contribution of induction versus NMD status was quantified by a five-fold cross-validated area under the ROC curve[16] for the full model, an induction-only model, an all-features-without-induction model, and an NMD-status-only model. Over-representation of productive-up and productive-down genes was tested with topGO[17] using the elim algorithm and Fisher's exact test and with one-sided Fisher tests against curated MSigDB Hallmark and Gene Ontology sets, using all genes passing the productive-analysis filter as the background.

Productive Response Change Level vs Composition

To separate change in total transcript output from change in isoform composition, the raw productive CPM change, ΔP, was decomposed following the method proposed by Kitagawa [18] into a level (output) term and a composition (isoform-mix) term:

$$\Delta f \ = \ f^{\ SMG1i} - f^{\ DMSO}$$

$$\Delta P \ = \ prodCPM^{\ SMG1i} - prodCPM^{\ DMSO}$$

$$\Delta T \ = \ CPM^{\ SMG1i} - CPM^{\ DMSO}$$

$$\Delta P \ = \overline{f} \times \ \Delta T \ + \overline{T} \times \ \Delta f$$

where $\overline{f}$ and $\overline{T}$ represent the midpoint between raw SMG1i and DMSO; the additive identity was confirmed numerically. Each gene was assigned to the level- or composition-dominated class by the larger of the two terms in absolute value (|level| vs |composition|). This classification was computed on NMD-target genes with a significant productive direction call (lfsr < 0.05) and reported separately for productive-up and productive-down genes. Non-NMD and uncertain genes, which lack an unproductive compartment and therefore have a composition term of zero by construction, were excluded.

The same causal distinction was found using the custom normalization. Significant productive up and down genes (AT2, LAE, MV) were partitioned by mechanism using the short-read gene induction L, the short-read gene-level log2 fold change, and the productive-fraction change Δf: transcriptional repression when L < −0.2 (total transcription fell, productive isoforms decreasing in parallel), composition shift when Δf < −0.10 with L ≥ −0.1 (transcripts redistributed into NMD-targeted isoforms with little change in total transcription), and mixed otherwise. Because L is a short-read gene-level estimate not derived from the long-read library, this classification provides a transcriptional signal independent of the isoform-level data used in the decomposition. The functional themes of the level and composition groups were compared by one-sided Fisher tests against curated Gene Ontology and Hallmark sets and by topGO over-representation against the same background. Example genes (*SHMT2, SRSF2, PCNA*) were rendered at isoform resolution with Isopair.

RNA-binding Protein and SR Protein Enrichment Among NMD Targets

To test whether NMD preferentially targets RNA-binding proteins (RBPs), we defined an NMD-target gene as any gene with at least one NMD susceptible isoform (long-read isoform-level mashr lfsr < 0.05 and posterior mean log2FC > 0) in at least one cell type, with the background comprising all genes possessing an isoform in the long-read DIE/mashr set. NMD-target genes were intersected with the Gerstberger, Hafner & Tuschl (2014) census of 1,542 human RBPs, matched on Ensembl gene identifier or HGNC symbol, and enrichment was assessed by one-sided Fisher's exact test against the background, both pooled and per cell type. As an orthogonal, experimentally defined reference, the test was repeated against the ENCODE eCLIP roster of RNA-binding proteins (Van Nostrand et al. 2020). Within each census we additionally tested enrichment of individual RBP functional classes (including splicing regulation, core spliceosome, exon-junction complex, 3′-end processing, and RNA stability) among NMD targets, using the same one-sided Fisher framework against the DIE-tested background.

The canonical serine/arginine-rich splicing factors (*SRSF1*–*SRSF12*) were checked for NMD susceptibility across the four cell types. To determine the splicing event responsible for the NMD susceptibility of each SR protein, we applied the Isopair framework (described below) to the canonical SR-protein genes *SRSF1*–*SRSF12*. For each SR gene carrying an NMD susceptible isoform, we paired its most highly expressed non-NMD isoform in DMSO (the reference) with its most highly expressed NMD susceptible isoform in SMG1i (the comparator) and enumerated the splice events distinguishing them. The NMD-triggering event was classified as a poison cassette exon, an intron-retention event (including alternatively spliced 3′-UTR introns), or loss of an internal exon, by intersecting each event's genomic coordinates with the comparator's premature termination codon. Premature termination codons were called under the canonical 50-nt rule from each isoform's primary open reading frame. Candidate ORFs were enumerated by a Kozak-aware scan (Isopair::enumerateOrfs), and the primary ORF was selected (Isopair::selectPrimaryOrf) by reference-AUG tracing – projecting the start codon of the gene's dominant non-NMD isoform onto the comparator's spliced sequence and, when that start codon was present, reading the ORF from it; where the reference start codon was absent, the highest-Kozak-scoring ORF was used. Kozak context (the −3 purine and +4 G determinants, scored as a position-weight log-odds) was used to adjudicate between competing start codons, in particular to compare the reference-derived start with the TransDecoder-predicted CDS. This procedure recovers occult PTCs that length-biased CDS predictors miss by selecting a downstream, in-frame start that bypasses the true (PTC-generating) initiator.

Isoform Pairs Analysis

In order to link splicing events to NMD susceptibility, we performed an analysis in which we focused only on annotated protein coding genes with at least three expressed isoforms (≥5% of overall gene expression in either DMSO or SMG1i and ≥5 reads in ≥1 sample) that met the following criteria: 1) the most highly expressed isoform in the gene was not susceptible to NMD, 2) the gene contained an expressed NMD susceptible isoform, 3) it contained at least one other expressed non-NMD isoform. For these genes, we then analyzed these three isoforms so that we could identify the splicing events that differentiated isoforms 2) and 3) from the dominant isoform, using this as a proxy for the splicing events that might have been responsible for the creation of those isoforms. This design ensures that when we compare NMD-related splicing transitions to Control splicing transitions, these comparisons share an identical reference isoform. We developed a suite of analytic functions contained in a package called Isopair which is publicly available on Github (https://github.com/peter4244/Isopair). Non-NMD susceptible isoforms were defined as those with mashr lfsr ≥ 0.05 and per-cell-type adj.P.Val > 0.50 in the limma-voom step underlying mashr.

Within each pair, Isopair detects the splicing events that distinguish the comparator from its reference. Twelve event categories are enumerated (skipped exon, 5′ alternative splice site, 3′ alternative splice site, intron retention, intron-retention 5′-difference, intron-retention 3′-difference, partial intron retention 5′, partial intron retention 3′, alternative transcription start site, alternative transcription end site, missing internal exon block, and a residual "other" category), and correctness is verified by the ability to achieve perfect reconstruction of the reference exon structure from the comparator.

For PTC analysis, comparator stop-codon position is intersected with splice-junction structure to determine PTC status under the canonical 50-nt rule (a stop codon is a PTC if it falls >50 nt upstream of the last exon-exon junction, producing one or more downstream EJCs after splicing). Three nested cohort scopes are used downstream: a strict all-annotated scope in which the three isoforms of the triplet are all GENCODE-annotated coding transcripts (n = 190); a broader ref-AUG-traceable scope in which the reference is annotated and its start codon can be projected onto each comparator's spliced sequence, re-intersected on (gene, reference) so the NMD and Control arms share the same gene set (n = 1,166); and a hidden-PTC subset of the ref-AUG-traceable scope in which the standard CDS caller does not flag a premature stop that the reference-anchored analysis reveals (n = 492). Because most NMD susceptible isoforms are novel, CDS regions are taken first from SQANTI3/TransDecoder (TD2) predictions and then, for isoforms classified as PTC-negative, recomputed from the reference isoform's annotated start codon ("ref-AUG tracing") to detect occult PTCs introduced by splicing-induced frameshifts. Splice events are then attributed as causal for PTC introduction by tracing which splicing event introduces or shifts the stop codon or terminal EJC. PTC-causing events are further classified into three mutually exclusive mechanism categories — frameshift (length change not a multiple of three nucleotides), in-frame stop (the event preserves the reading frame but introduces a new in-frame stop), and 3′ UTR splice (the event removes sequence between the natural stop and the terminal exon-exon junction).

Deep Learning Model

We trained an attention-based deep learning model to predict NMD responsiveness from transcript sequence and ORF-level structural features. Data were split into train/validation/test sets reserving chromosomes 2 and 4 for validation, chromosomes 1, 3, 5, and 7 for test, and the remaining autosomes plus sex chromosomes for training. Paralogous genes excluded from the test set.

Each transcript is represented as up to K=5 candidate ORFs, selected by priority: (1) the reference CDS ORF when the gene's dominant non-NMD isoform's start codon is present in the target isoform; (2) the SQANTI3/TransDecoder2 (TD2) CDS ORF when different from the reference; (3) the remaining slots filled by the top Kozak-scoring ORFs from a Kozak-aware

ORFik[9] scan. This priority scheme allows the model to choose between likely ORFs when determining NMD susceptibility.

For each of the K=5 ORFs, the model receives two 500-nt sequence windows (centered on the middle nucleotide of the AUG and stop codon, respectively) encoded with 9 channels per position - A/C/G/T one-hot, exon-exon junction position, rolling 50-bp GC fraction, and three reading-frame one-hot encoded channels relative to the ORF's AUG - together with five per-ORF structural features. These structural features are: location of ORF start as a proportion of total transcript length (frac_start where an AUG at the 5' most position would have a value of 0), the same type of encoding for stop position (frac_stop), binary indicator for whether ORF is the same as the reference isoform (is_ref_cds), binary indicator for whether ORF was the TD2 called ORF (is_sqanti_cds), and the number of splice junctions located downstream of the stop codon (n_downstream_ejc). Two shared-weight CNN branches encode the AUG and stop windows; a third linear branch encodes the structural features; the three sub-embeddings are fused into a per-ORF embedding and aggregated across the 5 ORFs by an attention layer whose weights are interpretable as ribosome ORF selection probabilities. A small classification head maps the transcript embedding to the NMD prediction. The model was trained using binary cross-entropy (BCEWithLogitsLoss) with a pos_weight derived from training-set class frequencies (n_neg / n_pos) to account for class imbalance. Optimization was performed via the Adam optimizer (initial learning rate 1e-3) with differential weight decay (1e-3 for the AUG and stop window CNN parameters; 1e-4 for all other parameters), a batch size of 256, and PyTorch automatic mixed precision (fp16). The learning rate was dynamically reduced upon reaching plateaus in validation AUC (ReduceLROnPlateau, factor 0.5, patience 5), and training was terminated using early stopping based on the same metric (patience 10) for a maximum of 100 epochs, using a fixed random seed of 42. We swept twelve combinations of AUG window size ({100, 500, 1000} positions) × stop window size ({100, 500, 1000, 2000} positions) and selected the AUG=500 / stop=500 configuration by held-out AUC.

For interpretability we used three complementary attribution methods. Joint DeepSHAP[10] was used to evaluate the relative contribution of the start and stop sequence windows along with the matrices of ORF-related features. This procedure varied the AUG window + stop window + 5 structural features associated with the ORF-0 (most likely ORF) simultaneously while holding ranks 1–4 fixed at observed values, yielding per-position per-channel sequence attributions and per-feature structural attributions for every test transcript. We ran 5 independent replicates with different random seeds (500 background transcripts each) and averaged the resulting attributions. To evaluate the importance of individual features within these three main groups of features, KernelSHAP at the fusion layer computed the exact additive Shapley decomposition of the prediction across the three sub-encoders (AUG branch, stop branch, structural branch). Subgroup-specific attributions were computed by stratifying the pooled DeepSHAP outputs by the mechanistic NMD subgroups defined in the results section. For occult-PTC validation we additionally enumerated ORFs with the ORFik package via Isopair::enumerateOrfs(), which performs an exhaustive Kozak-aware ORF scan independent of splicing context. ORF agreement between TD2, ORFik, and the reference AUG was assessed by start- and stop-codon coordinate match. Candidate ORFs were classified as credible upstream open reading frames if they simultaneously satisfied three structural criteria: open-reading-frame length below 200 nucleotides, both start and stop codons within the 5′ half of the transcript, and a Kozak position-weight-matrix score ≥ 1. Per-isoform attention weights from the aggregator layer were extracted to characterize the model's ORF-selection behavior, and within-class stop-codon frequencies at the priority-ORF stop codon were tabulated directly from the test set.

To benchmark our model against previously developed variant-level NMD predictors, we ran a head-to-head comparison against NMDetective-B[19] and the NMDEP-re-thresholded rule baseline[20] on the n = 1,166 ref-AUG-traceable cohort (1,050 NMD+/PTC+ + 116 NMD+/PTC− + 1,166 matched Controls; 2,332 isoforms total). The continuous gold standard for every isoform was the mashr posterior-mean log fold-change under SMG1 inhibition, aggregated across the four manuscript cell types (AT, DD, FB, MV) by taking the maximum per-isoform value. For each isoform we computed the four NMDetective-B features — in-last-exon membership of the (reference-AUG-projected, for PTC+/PTC−; natural, for Controls) stop codon, transcript-coordinate distance from start to stop, length of the exon containing the stop, and within-50-nt-of-last-EJ flag — directly from the long-read-observed transcript structure rather than by canonical-transcript lookup. NMDetective-B was scored with the published thresholds 50 / 150 / 407 nt and the leaf-level NMD-efficacy values (last-exon 0.00, start-proximal 0.12, long-exon 0.41, 50nt-rule 0.20, trigger 0.65 from Figure 1c of Lindeboom et al.); the NMDEP rule baseline used the same decision tree with NMDEP's grid-search-optimized thresholds 49 / 120 / 355 nt (Table 3 from Saadat et al.) and identical leaf values. The head-to-head intersection of isoforms scored by all three models was 2,218 (1,016 PTC+ + 95 PTC− + 1,107 Controls). For each model × stratum (pooled, NMD+/PTC+, NMD+/PTC−, Control) we computed Spearman correlation, Pearson correlation, R^2, mean absolute error, and root mean squared error against the gold-standard log fold-change. Because our model's predictions include training-set isoforms, all within-subclass metrics were additionally recomputed on the chr1 / chr3 / chr5 / chr7 held-out test split alone (n = 561; 255 PTC+ + 30 PTC− + 276 Controls) as a sensitivity analysis to confirm that the qualitative patterns were not artifacts of train-set inflation. Full-rank NMDetective-A and the full NMDEP MLP were deferred from this comparison because the full models are not publicly available.

Results:

Section 1: Isoform Discovery via Long-read RNA-sequencing

We sequenced 26 samples to an average depth of 13 million aligned FLNC reads, generating 341,638,920 total reads and identifying 162,800 expressed isoforms across 18,270 genes. Isoform length distributions were unimodal and centered near 2.5 kb, with minimal variation across cell types or treatments **(SF1)**. SQANTI analysis categorized 53.7% of transcripts as either full splice match (FSM - 55,770 isoforms) or incomplete splice match (ISM - 31,667 isoforms) to the GENCODE version 49 GTF; 44.4% of isoforms were novel (44,193 NIC and 28,045 NNC) with 1.9% falling into other transcript classes, including fusion and genic isoforms **(SF2)**. The median gene expressed five isoforms (mean 8.9, IQR 2–13, max 97), with 13,941 (76.3%) genes expressing two or more isoforms and 5,963 (32.6%) expressing ten or more.

To assess the agreement between short-read and long-read RNA-sequencing data, we aggregated long-read isoform counts to the gene level and filtered to a common set of 29,185 genes across 26 matched samples. Sample-wise concordance was uniformly high across all 26 samples, with mean Pearson r = 0.90 (range 0.83–0.91) and mean Spearman ρ = 0.89 (range 0.85–0.90) **(Fig 1A)**. Per cell type Pearson means ranged from 0.88 (FB) to 0.90 (AT2) **(SF3)**. Long-read and short-read quantification are therefore highly concordant on a per-sample basis, regardless of donor, treatment, or cell type, validating the downstream use of both platforms.

To examine how similar the isoform expression profiles were across cell types, we quantified cell type-specific expression in two ways: through differential expression analysis (difference in expression level between cell types) and identification of cell type-restricted expression (presence of absence of expression in DMSO samples). Among 105,938 isoforms expressed at testable levels under DMSO, differential expression analysis by cell type identified 28,930 isoforms significantly associated at 5% FDR with LAE, 22,131 associated to AT2, 21,429 associated to MV, and 18,422 associated to FB. Spearman correlations using isoform log-CPM values showed that FB and MV were strongly correlated (ρ = 0.79) while AT2 and LAE showed a more modest correlation with other cell types (ρ = 0.43-0.56) **(SF4)**. Despite the large number of significant associations with cell type, cell type-restricted expression was uncommon (0.36–2.24% of each cell type's expressed isoforms, **Fig 1B**). LAE contained the largest set of single cell type isoforms (8,087), followed by AT2 (2,267), MV (2,015), and FB (1,131). Only 27.8% (LAE) to 51.7% (MV) of cell type-restricted isoforms passed the additional expression filter required for differential testing, but among those that did all reached significance. Pairwise Jaccard indices based on binary isoform detection also showed extensive sharing of transcript repertoires with uniformly high values across cell types (0.86–0.92) **(SF4)**. Overall sharing is high when considering binary expression of isoforms, but in terms of expression level there are meaningful differences between cell types.

<u>Section 2: NMD Response</u>

To characterize the impact of NMD inhibition, we performed differential expression analysis at both the gene and isoform level. Features with local false sign rate (lfsr) < 0.05 and posterior mean logFC > 0 were classified as NMD susceptible, indicating increased abundance following SMG1i treatment and therefore basal degradation by NMD. At the short-read gene level, analysis of 25,955 genes per cell type identified 3,122–6,753 significant genes, of which 49–67% were NMD susceptible **(Table 1)**. At the long-read isoform level, analysis of 162,800 isoforms identified 24,803–35,336 significant isoforms per cell type, of which 90.9–92.0% were NMD susceptible **(Table 2)**. Across all cell types, 34,387 unique isoforms were NMD susceptible, 19,803 (57.6%) were core targets shared across all cell types, while 9,161 (26.6%) were cell type-specific, again with a heavy concentration in LAE (83.4%) **(Fig 1C)**. The mean posterior logFC of NMD susceptible features was substantially larger at the long-read isoform level than at the short-read gene level (**Fig 1D**), an expected result since NMD degrades specific transcript isoforms rather than entire genes. This demonstrates the benefit of long-read sequencing for studying NMD, which avoids signal dilution at the gene level arising from aggregation of NMD susceptible and non-susceptible isoforms of the same gene.

For NMD susceptible isoforms, we evaluated their expression levels and proportion of total gene expression arising from these isoforms, and we observed that these isoforms were concentrated at low isoform proportions – the majority fell below 50% of parent gene expression and a substantial fraction fell below 10%. However, these isoforms spanned more than four orders of magnitude of absolute expression **(SF5)**. Comparing the same isoforms between DMSO and SMG1i conditions showed a clear upward shift in both proportion and absolute expression, confirming that the low-proportion baseline is due to ongoing NMD-mediated degradation rather than constitutively low transcription. The fact that NMD susceptible isoforms cover a broad range of expression even in the DMSO condition indicates that NMD “tunes” the abundance of its targets rather than acting as a binary off-switch.

To quantify the extent of cell type-specificity in the NMD response, we computed two complementary measures from the mashr posterior distributions: per-cell type specificity (the fraction of NMD susceptible features significant in only one cell type) and pairwise effect sharing (the fraction of features significant in either cell type that shared both effect direction and magnitude within two-fold). At the isoform level, per-cell type specificity was higher in LAE (23.7%) compared to AT2, FB, and MV (0.8-3.4%), with the same pattern at gene resolution **(Fig 1E)**. Pairwise sharing was lowest for comparisons involving LAE (51-70% gene level; 35-72% isoform level), whereas the AT2/FB/MV trio showed substantially higher sharing (83-90% gene-level; 68-83% isoform-level) **(SF6)**. Together, these results indicate that while cell type-specific NMD targets can be identified, the predominant pattern is broad sharing of NMD effects across these four cell types.

The sharing of NMD effects we observed across lung cell types contrasts with a recent report by Tan et al.[13] that concluded that NMD susceptible mRNAs are predominantly cell type-specific, finding that only 16-25% of UPF2-dependent NMD susceptible transcripts were shared between human embryonic stem cells (hESCs) and neural progenitor cells (NPCs) (176 shared of 709 hESC and 1,088 NPC targets). We reasoned that the discrepancy could arise from how targets are compared across conditions: Tan et al. defined NMD targets by applying independent per-condition significance and fold-change thresholds in each cell type, a different approach than our mashr approach that explicitly models sharing between cell types. Accordingly, we reanalyzed the Tan et al. RNA-seq dataset using the same mashr-based approach used throughout this study (lfsr < 0.05 and positive posterior mean). Under this analysis, the transcript-level overlap of UPF2-dependent NMD targets between hESCs and NPCs increased to 41–50% (3,069 shared transcripts; 49.8% of hESC and 40.6% of NPC targets) with concordant effect direction observed across the two cell types for the large majority of shared targets. Thus, mashr-based reanalysis of the same data reveals substantially greater cross-context sharing of NMD targets and general agreement between studies.

Gene set enrichment analysis on the gene-level results (FDR < 0.05, NES > 0) showed a shared core of pathway targets with additional cell type-specific pathway enrichments (**Table 3; ST 1**). We identified five pathways significant in three of the four cell types (LAE, FB, MV; FDR 0.002–0.048; AT2 trended in the same direction at NES 1.6–1.8 but did not cross the FDR cutoff for these terms): cellular response to topologically incorrect protein, cellular response to unfolded protein, response to topologically incorrect protein, intrinsic apoptotic signaling in response to ER stress, and Reactome unfolded protein response. These pathways define a coherent ER-stress and integrated stress response program centered on the canonical PERK–ATF4–CHOP axis, with 75 unique leading-edge genes converging strongly across pathways on *ATF3*, *ATF4*, *DDIT3*, *DNAJB9*, *EIF2AK3*, *ERN1*, *HERPUD1*, and *HSPA5* across all responsive cell types. NMD therefore suppresses the integrated stress response under basal conditions in primary lung cells **(Fig 1F)**.

To examine pathway enrichment at the isoform level, we repeated the enrichment analysis on the long-read isoform-level results, scoring each gene by its largest-effect, significant NMD susceptible isoform. This isoform-resolved analysis recovered a large coherent shared program with 195 pathways significant in all four cell types (FDR < 0.05) **(Table 4; ST 2)**. The most consistently and strongly enriched was RNA splicing via the spliceosome (NES 1.29–1.40, FDR < 10^{-3} across all four cell types). Alongside this splicing program, the shared core spanned several further NMD-regulated programs. Cellular stress responses were prominent, encompassing the ER-stress and unfolded-protein response (NES 1.23–1.25, FDR ≤ 2×10^{-2}), the ATF4 amino-acid arm (amino-acid metabolism, NES 1.22–1.28, FDR ≤ 10^{-3}; tRNA aminoacylation, NES 1.35–1.43, FDR ≤ 10^{-2}), and a redox/antioxidant arm (oxidative-stress response, NES 1.12–1.15, FDR ≤ 10^{-2}). A second axis was protein and RNA homeostasis with the ubiquitin–proteasome system (NES 1.39–1.43, FDR ≤ 10^{-2}), ribosome biogenesis (NES 1.26–1.34, FDR ≤ 10^{-3}), cytoplasmic translation (NES 1.16–1.24, FDR ≤ 3×10^{-2}), and mRNA export (NES 1.24–1.36, FDR ≤ 3×10^{-2}).

The pathway enrichment prompted us to ask whether NMD broadly targets RNA-binding proteins, RBPs. NMD-target genes were strongly enriched for RBPs listed in the Gerstberger census (984 of 1542 were NMD targets; OR 3.3, $p < 10^{-3}$), and even stronger enrichment was observed when we used the orthogonal and more conservative ENCODE eCLIP roster of experimentally characterized RBPs (255 of 356 were NMD targets; OR 4.1, $p < 10^{-3}$; **SF7)**. Among RBP functional classes listed in the Gerstberger census, splicing regulators and core spliceosome components were the most enriched (splicing regulation OR 4.0; spliceosome OR 6.9), indicating that NMD preferentially regulates the splicing machinery itself **(ST 3)**. Given the well-described poison exon mechanism of NMD regulation for SR proteins, we closely examined NMD and splicing in all of the SR family, observing that 11 of 12 SRSF genes carried an NMD susceptible isoform. Across these 11 SR-protein genes, the NMD-triggering event was a poison cassette exon in 6 (*SRSF3*, *SRSF5*, *SRSF6*, *SRSF7*, *SRSF9*, *SRSF11*), intron retention in 4 (*SRSF1*, *SRSF2*, *SRSF8*, *SRSF10*), and gain of an internal exon in 1 (*SRSF4*) **(SF8-19)**. In *SRSF2* and *SRSF10* a cassette exon was gained but the premature stop arose upstream of it, so the exon inclusion event is not the NMD trigger.

Beyond the shared core, each cell type harbored a set of cell type-specific enriched pathways **(ST1; ST2)**. LAE harbored by far the largest cell type-specific program, spanning cilium movement and cilium- or flagellum-dependent cell motility, epithelial differentiation (keratinization and cornified-envelope formation), glycolysis, the HSF1-mediated heat-shock response, and nucleocytoplasmic transport. AT2-specific pathways centered on translation-regulatory programs and lipid and fatty-acid metabolism (fatty-acid oxidation, acyl-CoA and arachidonate metabolism, and protein–lipid complex assembly). FB-specific pathways centered on energy metabolism (glucose, ketone, and selenoamino-acid metabolism, mitochondrial β-oxidation) and genome maintenance (telomere maintenance and recombinational repair). MV-specific pathways centered on amino-acid and glutamate transmembrane transport, mitochondrial protein import and inner-membrane organization, and endothelial functions including leukocyte transendothelial migration and collagen biosynthesis. Thus while the core NMD regulatory program is conserved across cell types, NMD additionally regulates certain pathways in a cell type-dependent manner.

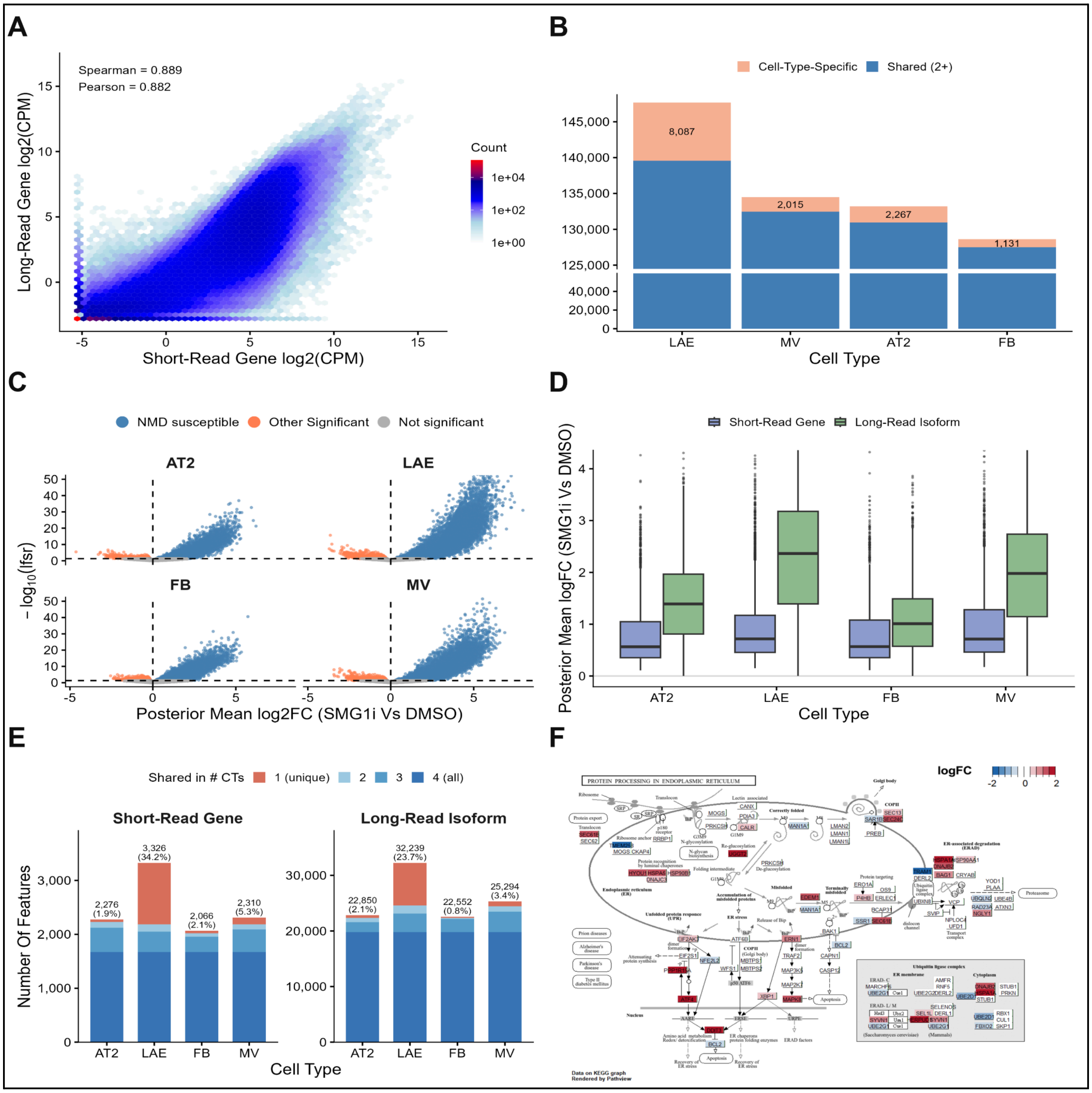

Figure 1 | Isoform Landscape and NMD Response. A) Sample-wise correlation of gene-level expression quantified by short-read and long-read RNA-seq. **B)** Number of expressed isoforms and cell type-restricted isoforms identified in each cell type under DMSO conditions **C)** Volcano plots showing differential isoform expression following SMG1i treatment in long-read RNA-seq data. NMD susceptible isoforms were defined as posterior mean logFC > 0 and lfsr < 0.05. **D)** Distribution of posterior mean logFC values for NMD susceptible features in short-read gene-level and long-read isoform-level analyses. **E)** Fraction of NMD susceptible genes and isoforms classified as cell type-specific across analysis platforms **F)** Gene level NMD susceptibility of the KEGG Protein Processing in Endoplasmic Reticulum pathway in LAE cells following NMD inhibition. Corresponding results for additional cell types are shown in SF19.

Section 3: Transcriptional Output Lost and Productive Isoform Response

To quantify the overall transcriptional output degraded by NMD, we quantified the "percent transcriptional output lost", defined as the summed positive expression differences (SMG1i minus DMSO) across all features divided by the total SMG1i expression. The measure was computed at both isoform and gene resolution. At the gene level, percent output lost ranged from 11.1% (AT2) to 12.5% (MV) **(SF 20)**. At the isoform level, percent output lost was larger, ranging from 14.4% (AT2) to 18.2% (LAE) **(Fig 2A)**. Within NMD susceptible isoforms, the per-isoform percent lost was broadly distributed (median 60-67%, IQR 30-100%, **SF21**).

We wanted to track the non-NMD isoforms of genes to uncover the coordinated processes that NMD regulates beyond the transcripts it directly degrades. To determine how the non-NMD isoforms of a gene (which we will refer to as the "productive fraction") respond when NMD susceptible isoforms accumulate, we measured the change in productive output under a custom normalization that removes library inflation in the SMG1i condition arising from the increased expression of NMD susceptible isoforms. This correction preserved gene rankings (Spearman ≥ 0.99 between the adjusted and uncorrected fold changes) and left non-NMD genes with closely matched expressions in DMSO and adjusted-SMG1i (Spearman ≥ 0.96; **SF22)**. Productive output was largely preserved: the median adjusted log2 fold change was near zero in every cell type (AT2 +0.14, LAE −0.04, FB +0.10, MV +0.05), with no significant change in roughly 83% of genes **(Fig 2B; ST4)**.

For the 17% of genes where we observed a significant change in the expression of the productive fraction, we applied a series of donor- and platform-based tests to confirm that these changes reflected reproducible signal rather than technical noise **(SF23)**. To ask whether the within-donor direction of change was more consistent than expected if the direction were random, we used a sign-flip permutation test, and we observed that the number of consistent effects far exceeded the permutation null in AT2, LAE, and MV (permutation $p \leq 10^{-3}$)[14]. To ask whether donors agreed on direction more often than chance, we compared cross-donor sign concordance to an exact binomial expectation, finding 1.2- to 2.6-fold more concordant genes than expected ($p \leq 10^{-3}$). Finally, to ask whether the variance of the observed effect-size distribution was wider than would be expected under the null, we compared the observed variance to the sign-flip permutation null (variance ratio 1.26–2.13). The signal was strongest in LAE, where Storey's π_0 estimator placed the fraction of genes carrying a real effect at 48%[15]. In FB, the signal was not robust across all of these evaluations, so we did not analyze those results further, but for the other three cell types all of the above evaluations suggested a clear signal of change in productive fraction for a subset of genes.

To characterize the change in productive output, we performed differential expression analysis with mashr on genes with and NMD susceptible isoform. Genes with lfsr < 0.05 and logFC > 0 were classified as upregulated and genes with lfsr < 0.05 and logFC < 0 were classified as downregulated. We found 88 (AT2) to 1337 (LAE) downregulated genes and 635 (MV) to 888 (LAE) upregulated genes **(ST4)**. Of the productive isoforms showing a significant response, there was upregulation in stress response pathways and downregulation in cell cycle and splicing regulation pathways **(ST4)**. Genes where productive output rose were enriched in the unfolded-protein response pathway (leading-edge genes including the ER chaperone *HSPA5* (BiP) and the UPR transducers *XBP1* and *EIF2AK3*) and for NF-κB/TNFα inflammatory signaling (HALLMARK TNFα via NF-κB, elevated in all cell types, $p \leq 10^{-2}$) **(ST4)** indicating activation of the ER-stress/integrated stress and inflammatory response upon NMD inhibition. This suggests that, while the 6 hour window seemed to capture mostly direct effects of SMG1 inhibition, there was some evidence of downstream effects of the NMD inhibition in selected pathways known to be regulated by NMD.

To distinguish how a gene's productive output changes under NMD inhibition, we applied a Kitagawa decomposition[18], which partitions the change in productive expression into two additive components: a level term, the change attributable to the whole gene being transcribed more or less, and a composition term, the change attributable to the gene's isoform balance shifting. Across all NMD susceptible genes with a significant productive change, the level term was dominant in 86.2% of genes and the composition term in only 13.5%, indicating that change in productive output is governed primarily by transcriptional level rather than by the redistribution of isoforms **(ST4)**. This split was strongly direction-dependent. Genes that gained productive output were almost entirely level-driven (99.2% level, 0.3% composition), consistent with increased transcription raising productive and NMD susceptible isoforms together. The composition term instead concentrated in the genes that lost productive output. Among significant productive-down NMD genes, 32.1% were composition-dominated, showing that a shift of the isoform pool toward NMD susceptible transcripts is a meaningful driver of productive loss specifically, yet a reduction in overall transcription (the level term, 67.9%) remained the more common cause. The same conclusion held under our custom NMD-anchored normalization, with short-read gene-level induction added as an external transcriptional signal not derived from the long-read library **(ST4)**.

Two distinct mechanisms partition all the productive-down genes. Composition-shift genes were specifically enriched for RNA splicing (OR 2.5, $p < 10^{-3}$) and SR/hnRNP factors (OR 6.8, $p < 10^{-3}$), together with snRNA-, rRNA- and mRNA-processing terms **(ST4)**. Notably, of the 12 canonical SR proteins, every significant change in productive output was a decrease (8 in LAE, 3 in AT2, 2 in MV) and none increased in any cell type. These factors regulate splicing transcriptome-wide, so their coordinated loss of productive output upon NMD inhibition could in turn propagate into broader splicing changes, consistent with prior reports that NMD shapes alternative splicing through splice-factor autoregulation[21]. Transcriptional-repression genes were instead enriched for DNA replication and repair (OR 2.1, $p < 10^{-3}$), the genome-maintenance program downregulated during cell-cycle arrest. The direct, isoform-level NMD mechanism acts predominantly on the RNA-processing machinery, whereas the larger transcriptional-repression class reflects the secondary cell-cycle arrest.

Three genes illustrate the mechanisms behind the opposing productive responses. In the up-arm, the direct *ATF4* target *SHMT2* was NMD susceptible yet significantly increased its productive output upon NMD inhibition (MV, 86.5 to 111.1 CPM; adjusted logFC +0.42), in step with the overall gene induction (short-read logFC +0.56; **Fig 2C)**. *ATF4* is NMD susceptible, so inhibiting NMD de-represses *ATF4* and activates its transcriptional program which causes induction of *SHMT2* raising its productive output. The down-arm comprises two distinct mechanisms. The first is autoregulation of the splicing machinery which *SRSF2* illustrates directly. As its NMD susceptible isoform (generated by a spliced 3′-UTR intron) surged, its productive output fell (84 to 50 CPM; adjusted logFC −0.73) and its productive fraction collapsed from 91% to 26% **(Fig 2D)**. The second is transcriptional repression which *PCNA* shows. *PCNA* lost roughly half its productive output (LAE, 251 to 130 CPM; adjusted logFC −0.95), but its productive fraction moved only slightly (100% to 93%) and its short-read transcription fell in parallel (logFC −0.81), indicating the whole gene was transcribed less rather than its isoforms redistributed **(Fig 2E)**.

We wanted to see if the productive response extended to genes without an NMD susceptible isoform which we term non-NMD genes. Running the same analysis, we found 36 (AT2) to 408 (LAE) downregulated non-NMD genes and 160 (MV) to 252 (LAE) upregulated non-NMD genes. The NF-κB/TNFα inflammatory up-arm and the DNA repair down-arm were also present in non-NMD genes, indicating a cell-wide response rather than one confined to direct NMD targets. These genes have no NMD susceptible isoform, so their productive change must be a secondary, downstream consequence of NMD inhibition which is supported by the smaller number of significant genes.

To understand these significant changes in productive fraction, we asked two questions: 1) What determines whether a gene's productive output changes? 2) What determines the direction of that change? NMD susceptibility governed whether a gene responded at all. In one-vs-rest models it raised the odds of both an increase and a decrease by nearly identical amounts (odds ratios 1.85 and 1.89), marking a gene as responsive without setting the sign. For genes with a significant change in productive fraction, the direction of change was negatively correlated with the gene's level of baseline expression **(ST4)**. In other words, genes with high baseline expression level tended to show a decrease in productive fraction during SMG1i, and genes with low baseline expression levels showed an increase. Further, the decrease was concentrated in LAE, which was ~18-fold more likely than AT2 to lose productive output, consistent with it harboring the largest and most cell-type-specific NMD program. A cross-validated model predicted the sign of change from independent short-read gene logFC almost as well as from all features combined (AUC 0.93 vs 0.96), but barely from NMD status alone (0.54). NMD inhibition therefore imposes no direction-specific penalty, and productive output simply rises or falls with the gene's underlying transcription **(ST4)**.

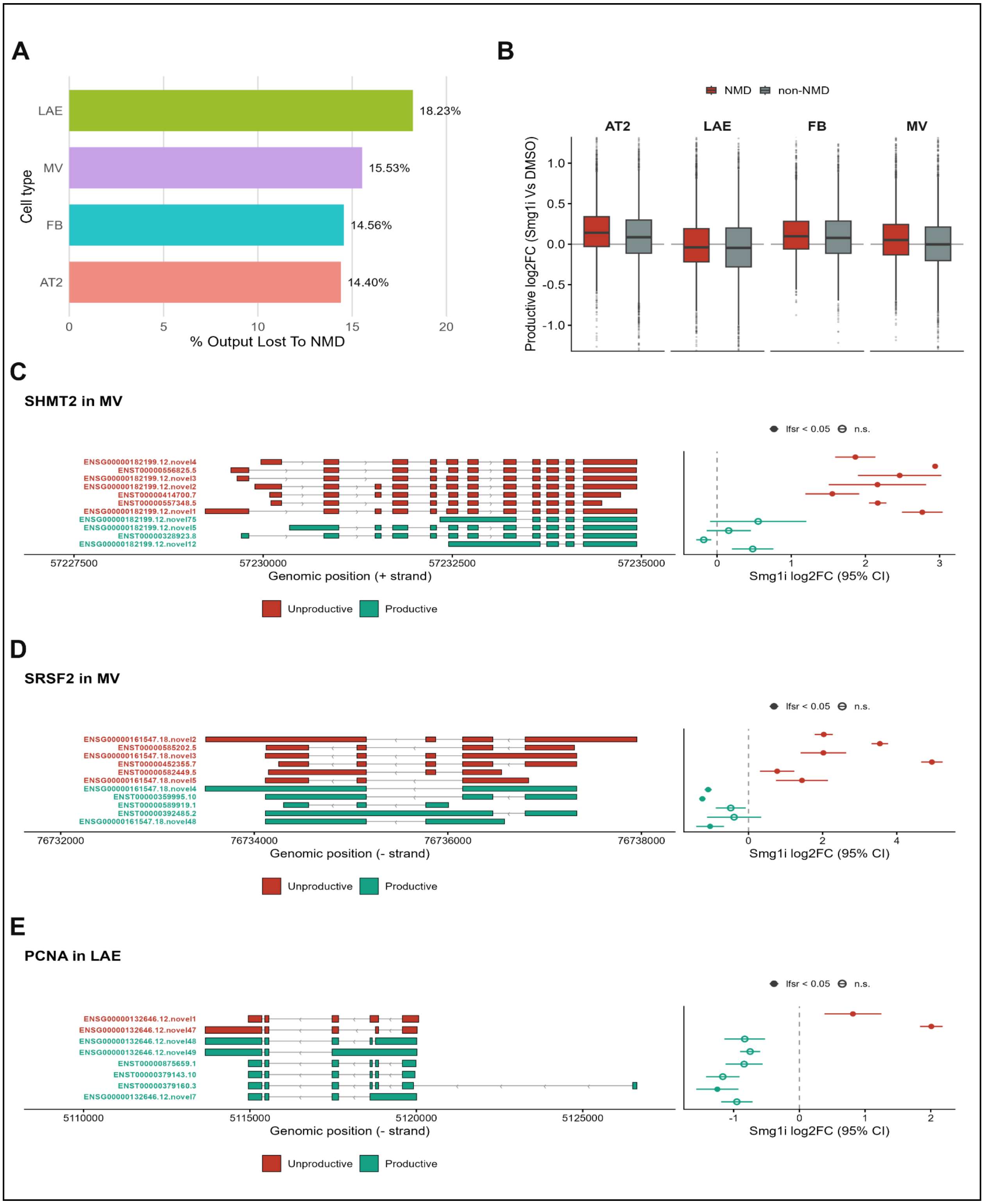

Figure 2 | Transcriptional Output Lost and Productive Isoform Response. A) Percent transcriptional output lost to NMD at the isoform level across cell types. **B)** Distribution of productive mean logFC values for NMD and non-NMD genes in all cell types. **C)** Example of an up-regulated productive fraction stress response gene (*SHMT2* in MV) showing structure and logFC of productive and unproductive isoforms rendered using Isopair. **D)** Example of a down-regulated productive fraction splicing gene (*SRSF2* in MV) showing structure and logFC of productive and unproductive isoforms rendered using Isopair. **E)** Example of a down-regulated productive fraction DNA repair gene (*PCNA* in LAE) showing structure and logFC of productive and unproductive isoforms rendered using Isopair. **Abbreviations:** NMD genes are genes with an NMD susceptible isoform; non-NMD genes are genes without an NMD susceptible isoform; Unproductive Isoforms are NMD susceptible isoforms; Productive isoforms are all isoforms that are not susceptible to NMD.

Section 4: Attributing NMD Susceptibility to Specific Splicing Events

In order to identify the splicing events that shift transcription towards NMD susceptible isoforms, we conducted analysis of pairs of isoforms to identify the splicing events that differentiate one isoform from another. The codebase and methods for this analysis are available as the Isopair package (https://github.com/peter4244/Isopair). In total, Isopair identifies 12 different categories of splice events (see **SF24 - Isopair Splice Events**), and correctness is verified by the ability to reconstruct the reference isoform completely given only the comparator isoform and the splice event list. First, we identified genes with at least 3 expressed isoforms where at least one isoform triggers NMD. This resulted in 3,009 genes from which the isoform pairs were drawn (illustrative example, **Fig 3A**). These genes had a median of 7 isoforms each (**SF25 - Isoform Count in Isoform Pair Sets**), and the median reference isoform accounted for 31% of its parent gene expression, with 38% of the reference isoforms accounting for >50% of parent gene expression. We then identify the “reference isoform”, i.e. the highest expressed non-NMD isoform, against which we can compare both the NMD susceptible isoform (the NMD comparison) and the next highest expressed non-NMD triggering isoform (the Control comparison). As expected, the NMD and non-NMD comparator isoforms were expressed at lower levels than the reference isoform, but both comparator isoforms were expressed at similar levels in the SMG1i samples (**SF26 - Expression Levels of Isoform Pair Sets**). Transcript length was similar for reference and NMD comparator isoforms (median length 2893 and 3049 nucleotides, respectively), and Control comparator isoforms were slightly shorter (median 2762 nucleotides, $p<0.001$ for comparison to both reference and NMD comparator, **SF27 - Transcript Length Comparison**). The flow chart documenting the construction of these isoform NMD and control pair sets is shown in (**SF28 - NMD pairs flowchart**).

By comparing these two sets of isoform pairs (NMD and control), the Isopair algorithm identified splicing events that differentiate NMD susceptible from non-susceptible isoforms within the same gene, and in many cases we could identify the specific splicing events that introduce PTCs to NMD susceptible transcripts. We first examined the overall sequence similarity of the NMD and Control comparator isoforms to the reference, noting that the NMD isoforms shared more sequence content with the reference, suggesting that NMD arises from relatively targeted splicing events rather than disordered splicing (**Fig 3B - Sequence Similarity of Isoform Pairs**). In terms of the frequency of splicing events, there was a marked increase in skipped exon events which were twice as frequent in NMD pairs compared to Controls (**Fig 3C - Splicing Event Prevalence**). Conversely, intron retention events were more common in Control pairs. We also tracked the balance of sequence gain versus loss, and we observed that terminal events (Alt TSS and TES) and skipped exons more frequently led to sequence gain in NMD isoforms, whereas intron retention events led to more sequence gain in controls (**SF29, Gain Direction by Event Type**).

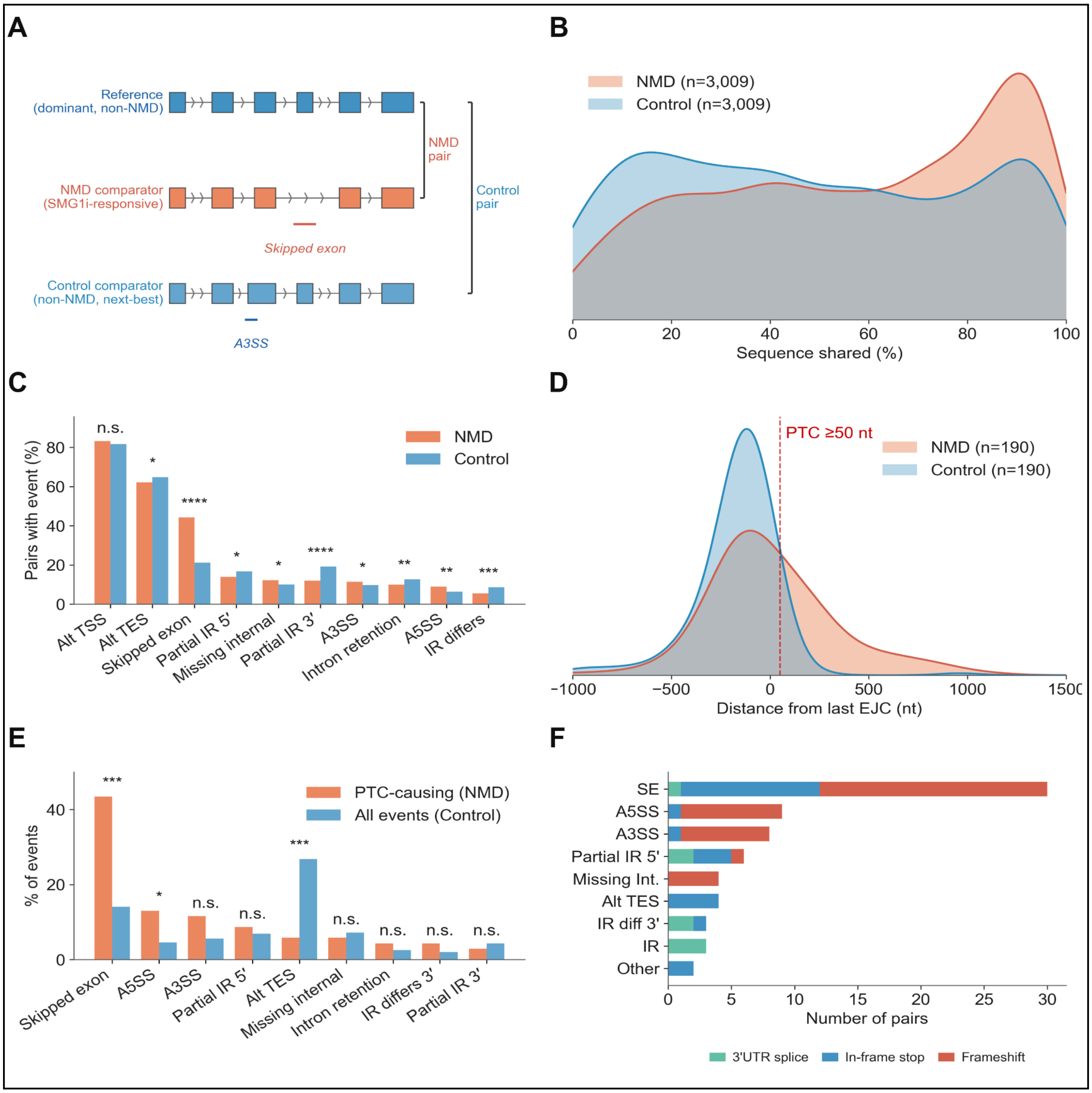

Figure 3 | Isopair analysis of NMD susceptible isoforms and splicing events. (A) Schematic of the NMD and Control pairs drawn from the same gene. (B) Distribution of sequence similarity between reference-NMD pairs and reference-Control pairs. (C) Per-event-type prevalence in pairs where all three isoforms are GENCODE annotated (n=190). SE is twice as prevalent in NMD pairs (44.2 vs 21.2%, p ≈ 10^{-81}). (D) Distance from each comparator's stop codon to its last exon–exon junction. Dashed line marks the 50-nt PTC threshold. Direct PTC rate 37.9% NMD (72/190) vs 2.1% Control (4/190). (E) Proportion of each event type among Isopair-attributed PTC-causing events in NMD (coral; n = 69 attributed events from 72 PTC+ pairs) vs all events in 190 Controls (light blue; n = 447 total events). SE accounts for 43.5% of PTC-causing events vs 14.1% of Control events (Fisher $p = 8\times10^{-8}$); A5SS is enriched (13.0% vs 4.5%, $p = 9\times10^{-3}$); Alt TES is depleted (5.8% vs 26.8%, $p = 3\times10^{-5}$). (F) PTC-causing events stratified by mechanism: 69 attributed pairs split 38 frameshift (coral) / 23 in-frame stop (blue) / 8 3′UTR splice (teal) = 55% / 33% / 12%. ***p < 0.001, **p < 0.01, *p < 0.05, ns, non-significant (Fisher's exact). **Abbreviations.** SE, skipped exon; A5SS / A3SS, alternative 5′ / 3′ splice site; IR, intron retention; Partial IR 5′/3′, partial intron retention at the 5′/3′ end; IR differs, intron retained in both isoforms with different boundaries; Missing internal, internal exon present in reference but absent in comparator; Alt TSS / TES, alternative transcription start / end site.

We next examined NMD susceptible isoforms for the structural features reported to be associated with NMD - namely premature termination codons (PTCs), 3' UTR length, and presence of 5' upstream open reading frames (uORFs). However, each of these calculations depends on proper specification of the CDS, which in the case of novel isoforms (not present in GENCODE) was computationally derived from the TD2 program[22], which prioritizes longer ORFs and downweights PTC-containing isoforms. To avoid this potential bias, we analyzed only the isoform pairs where the reference, NMD comparator, and Control comparator are all present in GENCODE with an annotated CDS, yielding 190 NMD and Control pairs in which every isoform's PTC status can be evaluated directly from its own GENCODE-annotated stop codon. Within this scope, we observed an 18-fold enrichment of PTCs in NMD susceptible isoforms (37.9% vs 2.1% PTC rate, $p < 10^{-20}$). We observed a strong enrichment of stop codons far upstream of the terminal exon junction in NMD isoforms (**Fig 3D - PTCs in NMD**) with a clear dose-response relationship between upstream distance of the stop codon and the magnitude of the NMD response (**SF30 - PTC distance dose response**). There was a similar relationship with the number of downstream EJCs, where the magnitude of the NMD response peaked at 4-5 downstream EJCs from the PTC (**SF31 - NMD Effect Size by Number of EJCs)**. When we examined how PTCs were introduced to NMD isoforms, frameshift events were the leading mechanism (55%), followed by in-frame stop events (33%) and 3' UTR splicing (12%, which shifts the terminal EJC downstream of a normal stop codon). Skipped exon was the most common event type (44% of PTC-causing events vs 14% in Controls, Fisher $p < 10^{-7}$), with A5SS also significantly enriched (13% vs 4.5%, $p < 10^{-2}$) (**Fig 3E and F**).

We then focused our attention on the NMD+ isoforms in this subset for which a PTC was not identified (NMD+/PTC- isoforms), and we compared the following quantities between NMD+/PTC+, NMD+/PTC-, and Control isoforms: CDS length, 5' UTR and longest uORF length, and 3' UTR length. We observed that the NMD+/PTC- isoforms had longer 5' UTRs and uORFs than both other groups (**Figure 4A and B**), suggesting that NMD in NMD+/PTC- isoform was driven by the presence of uORFs. There was no significant difference in CDS or 3'UTR length between the NMD+/PTC+ and NMD+/PTC- isoforms (**SF32**). We then expanded our set of isoform pairs to include novel isoforms. Since the CDS in the NMD+ novel isoforms was called by TD2, which might be biased against PTC-containing isoforms, we compared the TD2-called CDS to the what the CDS would be if it used the same AUG that was used by the reference isoform in that gene, an approach we term “reference AUG tracing.” Limiting our isoform pairs to those where the reference AUG was contained within the comparator isoform yielded 1166 NMD and Control pairs. In 50% of these cases, the TD2-called CDS used the same AUG as the reference isoform CDS. When the CDS calls differed, the TD2 CDS was usually longer than the reference AUG-tranced CDS and used an AUG downstream from the reference AUG (**SF33**). This trend was much stronger in the 492 novel NMD+ isoforms where the ORF defined by the reference AUG included a PTC. In that case, 99% (487/492) of the TD2-called CDS were downstream from the reference AUG, demonstrating that when choosing a CDS in these cases TD2 skipped over the PTC-containing reference ORF in favor of a longer, non-PTC containing downstream ORF. To determine which of these ORFs had a stronger start codon, we calculated Kozak scores for the TD2 selected AUG and the reference AUG, and we observed that the reference AUG was stronger in 78% of cases (384/492, **SF34**). Collectively this demonstrates a negative bias for TD2 CDS calls against PTC-containing ORFs, so we calculated what the PTC rate in NMD+ isoforms would be if relied on reference AUG tracing to determine the CDS. With this approach, we identified PTCs in 90% of the NMD+ isoforms (1050/1166), still observing that the 5' UTR and longest uORF lengths were substantially higher in the NMD+/PTC- isoform subset (**Figure 4C and D**).

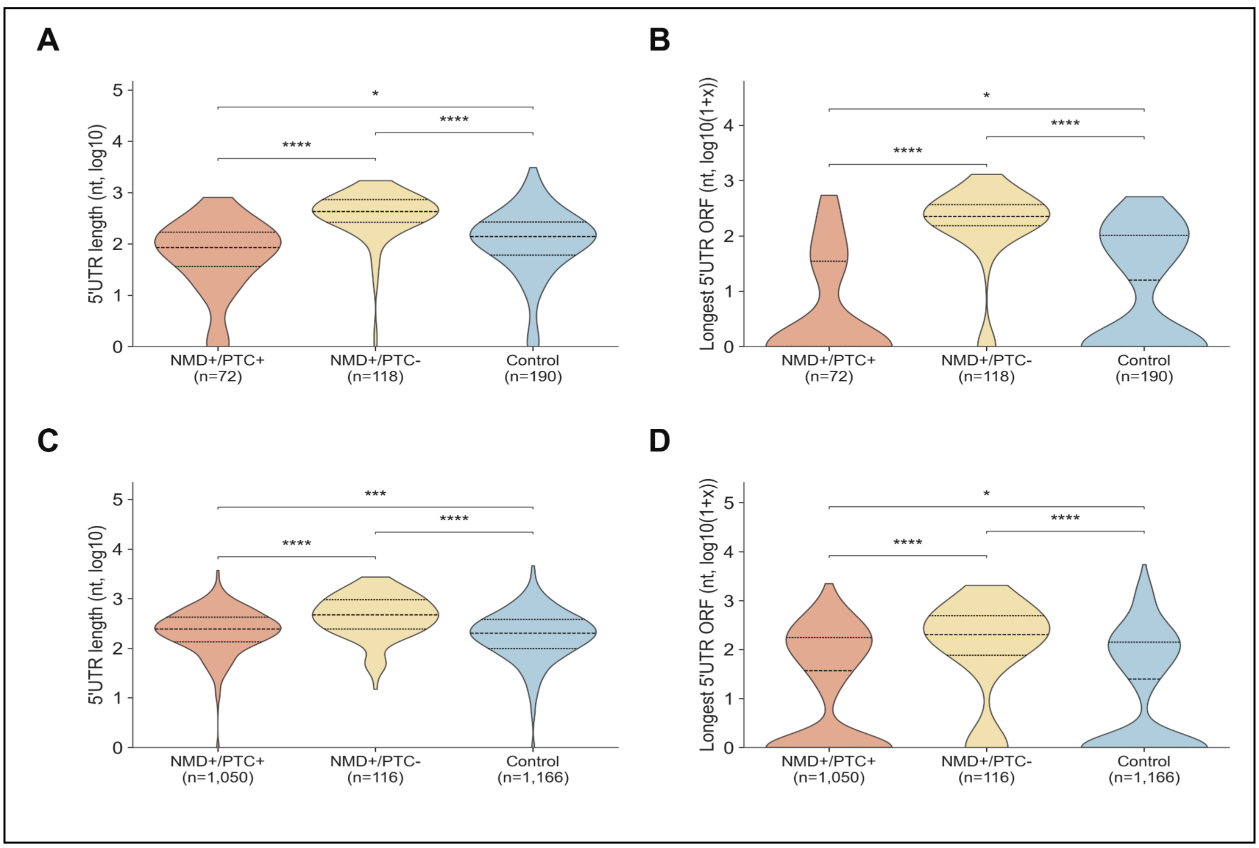

Figure 4 | Analysis of NMD+/PTC- isoform structure in Isopair pairs. (A) 5′UTR length in NMD+ isoforms with and without PTCs along with Control comparators from the same genes. Isoforms were selected only from genes where the reference, NMD+, and Control isoforms were all in GENCODE (n = 72 NMD+/PTC+, 118 NMD+/PTC−, 190 Control). Horizontal lines mark q25 (dotted), median (dashed), q75 (dotted). (B) Longest 5′UTR ORF length for the same set of isoforms. (C) 5′UTR length in a broader set of isoform pairs where only the reference isoform needs to be in GENCODE. Of n = 1,166 NMD comparator pairs in which the reference AUG could be traced through the comparator's spliced sequence, 1,050 (90%) had a downstream-EJC stop (NMD+/PTC+) and 116 had no downstream EJC (NMD+/PTC−); the 1,166 ref-AUG-traceable Control pairs are the matched non-NMD baseline. (D) Longest 5′UTR ORF length for the same set of isoforms. ***$p < 10^{-4}$, **$p < 10^{-3}$, *$p < 0.05$, n.s. (Wilcoxon with Hodges–Lehmann shift).

Since there is some controversy about the importance of 3’ UTR length as a driver of NMD in humans, we examined 3’ UTR length in NMD+/PTC+, NMD+/PTC-, and Control isoforms in both the GENCODE restricted isoform set (n=190) and the more expansive pair set (n=1166). There is another nuance in terms of calculating 3’ UTR length, which is that in PTC+ isoforms 3’ UTR length increases by virtue of the PTC, so we also calculated the 3’ UTR length in these isoforms from the first stop codon available within 50nt of the terminal EJC. Using this estimated “natural” 3’ UTR length, there was no significant difference in 3’ UTR length between NMD+/PTC+ isoforms and the other groups in the GENCODE restricted set (**SF35**). In the expanded set, NMD+/PTC+ isoforms had significantly longer isoforms than both other groups ($p<0.001$ - median length 1290/800/948 for NMD+/PTC+, NMD+/PTC-, and Control respectively).

Section 5: An Interpretable Predictive Model for NMD

Based on our Isopair analysis, the majority of NMD in our experiment was explainable by the presence of PTCs or features in the 5’ UTR. To investigate this further and identify sequence elements driving NMD susceptibility, we trained a deep learning model that predicted the likelihood of NMD from transcript sequence. The architecture of this model was designed to achieve both high predictive performance and interpretability specifically for sequence features near translational start and stop sites. The input data for the model were provided as an isoform sequence and structural information derived from up to five candidate CDS regions, including the relative coordinates of the start and stop codons within the transcript (frac_start and frac_stop), binary indicators of whether an ORF was identified as CDS in either GENCODE or by TD2, and the total count of downstream exon-exon junctions. The input data were structured as three distinct blocks of information related to translation START site, STOP site, and ORF structural data with the START and STOP blocks provided as raw sequence and the structural data provided as a tabular matrix (**Figure 5A**). These data were fused in an internal layer which fed into an attention layer that allowed us to identify which candidate CDS the model preferred for each isoform. For model evaluation, isoforms from non-paralogous genes on chromosomes 1,3,5, and 7 were held out.

These models could predict NMD with high accuracy. The overall best model used 500nt windows around both the start and stop sites, yielding an AUC of 0.93 and AUPRC of 0.83 (**Figure 5B**). Using Shapley value analysis, we observed that 60% of the predictive information came from the ORF structural data with the remaining 40% coming from the START and STOP sequence information. The STOP site sequence was nearly three times as important as the START sequence (**Figure 5C**). Of the individual ORF structural features, EJC count was by far the most important (**Figure 5D**).

To determine what sort of sequence features drove prediction, we plotted the Shapley values across the START and STOP sequence windows, observing the same pattern of increasing importance around the start and stop codons, with the most importance concentrated within the 50nt preceding the start or stop site (**SF36**). In both windows, the model places more importance on NMD than Control isoforms. Nucleotide level importance around both sites is shown in panels E and F. At the start site, the model learned Kozak sequence information that determines strength of the start site. There is no weight placed on the AUG itself since this sequence is invariant across all the ORFs provided in training. For the STOP window, the model places importance on which of the three stop codons is used, with UGA shifting predictions towards NMD. We confirmed that, when comparing NMD to non-NMD isoforms, UGA is more commonly used (56% vs 48%, $p<0.001$, **SF37**). The UGA codon is considered to be the "leakiest" codon that predisposes to translational readthrough, a trend that is also supported by the largest importance for U at the +4 position (first base after the stop codon) that is also associated with readthrough[23].

Our models also included an attention layer that integrates information from the candidate ORF/CDS regions provided for each isoform. The ORFs were numbered according to the CDS likelihood of each ORF according to our prioritization criteria (see Methods). Attention analysis demonstrated that, while most of the attention was focused on ORF0 (the most likely CDS), attention was more broadly distributed for NMD than Control isoforms (**SF38A and B**). To determine whether the model identified uORFs as important features for NMD+/PTC- isoforms, we identified uORFs (see Methods) and quantified the attention weights of uORFs in NMD+/PTC+, NMD+/PTC-, and Control isoforms from our gene matched isoform set (n=1166). This demonstrated that the model does upweight uORFs in NMD+/PTC- isoforms (**Fig 5G**). We also examined the Shapley scores for the START and STOP windows stratified by isoform PTC subclass, and we observed that the START window information is twice as important in the NMD+/PTC- than in the NMD+/PTC+ isoforms (**SF39**). However, overall predictive performance was substantially lower in NMD+/PTC- isoforms (**SF40**).

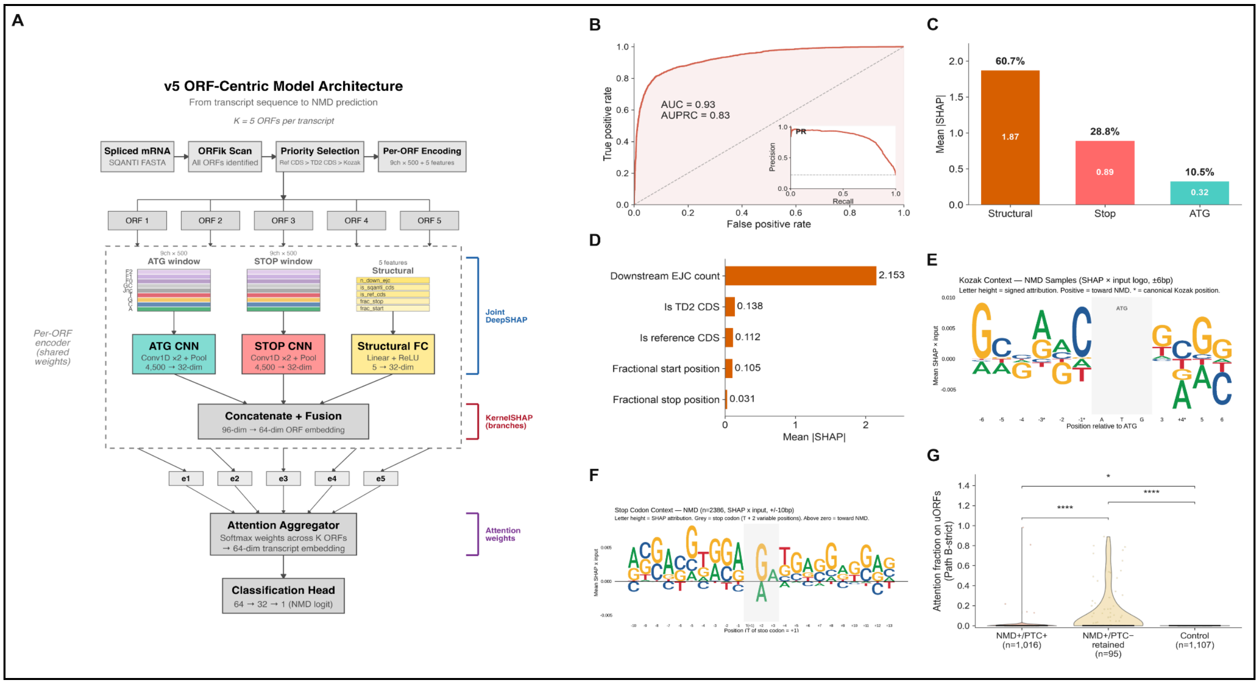

Figure 5 | A deep-learning model predicts NMD susceptibility from sequence and learns canonical start- and stop-codon context. (A) Model architecture. Per transcript, the ORFik scan and a priority selector pick five candidate ORFs (reference CDS > TD2 CDS > top-Kozak); each ORF is encoded as a 9-channel × 500-nt window around its AUG and its stop codon plus 5 structural features. Shared-weight CNN branches encode AUG and stop windows; a linear branch encodes structural features; concatenation and fusion yield a 64-dim per-ORF embedding. Softmax attention aggregates the five ORF embeddings into a single transcript embedding that feeds a classification head returning the NMD logit. Bracketed labels indicate where each interpretability method operates: Joint DeepSHAP at the input layer, KernelSHAP at the fusion layer, attention weights at the aggregator. (B) Held-out performance on the chr-1/3/5/7 paralog-free test set. AUC = 0.93, AUPRC = 0.83. Diagonal = chance; PR-curve inset (lower-right). (C) KernelSHAP branch decomposition at the fusion layer: Structural 60.7%, Stop 28.8%, AUG 10.5% of the total NMD-call signal. (D) Mean |SHAP| per structural feature (5 DeepSHAP replicates pooled). Downstream EJC count dominates (2.153), ~15× the next feature. (E) Per-nucleotide signed SHAP × input around the start codon, NMD class. Letter height = mean signed attribution; positive (above zero) = pushes prediction toward NMD. Canonical Kozak preferences (G/A at −3, C at −1, G at +4) are recovered as the dominant positive letters at those positions. (F) Same construction for the stop codon. (G) Fraction of the model's attention directed to candidate upstream ORFs (uORFs) in the NMD+/PTC+, NMD+/PTC-, and Control isoform subgroups. ****p < 10^{-10}, *p < 0.05 (Mann-Whitney U). **Abbreviations:** TD2, TransDecoder2; CNN, convolutional neural network; AUC, area under the ROC curve; AUPRC, area under the precision–recall curve; SHAP, SHapley Additive exPlanation; Kozak PWM, canonical Kozak position weight matrix.

One of the strongest signals is a difference in GC content between NMD and Control isoforms that is most prominent after the stop codon. This likely reflects the effect of PTCs, which effectively turn exons (which are GC-rich relative to UTRs) into 3' UTR sequence. To prevent the GC signals from overwhelming more region-specific signals like Kozak sequence, the model incorporates a channel specifically for GC content, and it is clear that GC content is a strong signal differentiating NMD from Control isoforms in the STOP but not the START window (**SF41**).

To benchmark our model's performance, we compared it to two other predictive models for NMD - NMDetective-B and the NMDEP rule-based model baseline, choosing for both the simpler rule-based models that could be replicated in our dataset. Focusing on isoforms from our gene-matched set in testing data only (n=561 isoforms from 293 genes), we compared predictions from each of the predictive models against the log2FC from the mashr analysis for NMD susceptibility, using the correlation between predicted and observed values from the test dataset as our evaluation metric. We observed that the three models had nearly identical performance in the entire dataset (Spearman rho = 0.76-0.77 for all three models). When we stratified isoforms by NMD and PTC status (NMD+/PTC+, NMD+/PTC-, and Control), we observed higher correlations with our model predictions, indicating that our model seems to have learned additional features beyond the explicit rules in the other two models for making NMD predictions (**SF42**).

Discussion

This study of NMD in primary lung cell types is the first transcriptome wide description of NMD with isoform-level characterization from long read sequencing, and it has a number of important findings. First, long-read RNA-seq is uniquely well-suited to study NMD since it is an isoform-level phenomenon, and long-read isoform-level analysis identifies stronger NMD effects than short-read sequencing. Second, NMD is widespread, involving thousands of genes in each cell type, and in each of our studied cell types alternative splicing was responsible for creating a pool of NMD susceptible transcripts in thousands of protein-coding genes. Third, NMD targets are enriched in multiple biological pathways, but especially those related to alternative splicing and cell stress responses. Fourth, while cell type-specific NMD targets can be identified, statistical analysis of effect sharing suggests that many NMD effects are shared across cell types. Finally, isoform-level data on NMD allows for the development of accurate sequence-based predictive models that can be used to develop and evaluate mechanistic hypotheses regarding NMD. Our analyses confirmed prior observations regarding the importance of PTCs and uORFs in NMD and the prominent role of exon skipping/inclusion as means of introducing PTCs. Our model also provided novel information about potential links between NMD-associated sequence features and propensity for translational readthrough.

This study is the first to our knowledge to apply long-read RNA-seq to identify targets of NMD, and it illustrates the benefits of isoform-level resolution for the study of NMD. Compared to short-read data, the effect sizes for long-read data were larger and more consistent with the effects of NMD inhibition. By generating short and long-read RNA-seq data from the same samples, we are able to make direct comparisons of the performance of each technology which reveals that long-read identifies more targets of NMD and allows for direct quantification of the productive and non-productive portions of total transcriptional output, as we discuss later.

Our study confirms previous reports that NMD is widespread. While Fair et al.[3] reported that 15% of the transcriptional output for protein coding genes is degraded by NMD, we confirmed a similar range of effects in our primary lung cell types, with slightly higher estimates obtained when we analyzed this at the isoform rather than gene level (14-18% of transcriptional output degraded). NMD is most frequently discussed in two different scenarios - the “rare mutation scenario” where PTCs arise from low frequency germline or somatic variants and the “AS-NMD” scenario where PTCs arise from the effects of alternative splicing on primarily non-varying sequence. Our data provide additional evidence of the widespread presence of AS-NMD, and our analysis reveals how AS-NMD broadly shapes the transcriptome while also showing significant enrichment in specific pathways.

Among the pathways enriched in NMD targets, the most common pathway enrichments observed across cell types were related to the cellular stress response and the regulation of RNA splicing. In our data, five overlapping ER-stress and unfolded protein response terms reached significance in three of four cell types, and leading edge analysis converged tightly on the canonical PERK–ATF4–CHOP axis (*ATF4*, *DDIT3*, *PPP1R15A*, *ATF3*, *CHAC1*). Cellular stress response has been previously linked to NMD, including the first transcriptome-wide profile of NMD substrates that identified stress-response factors such as *ATF3* and *ATF4* (Mendell et al. 2004). The core integrated stress response factors *ATF4*, *ATF3*, and *CHOP* are direct NMD substrates whose mRNAs accumulate when the pathway is suppressed (Gardner 2008; Karam et al. 2015), and this pathway exemplifies how uORF-mediated NMD susceptible transcripts may produce functional protein under certain conditions. In the case of *ATF4*, the master regulator of the integrated stress response, its 5′ uORFs both render the transcript NMD susceptible and gate its translation, since stress-induced eIF2α phosphorylation delays ribosomal reinitiation to permit *ATF4* synthesis (Vattem & Wek 2004). Quantitative proteomics has further shown this regulation is concentrated specifically on the PERK–ATF4 branch of the Unfolded Protein Response rather than distributed evenly across its three arms (Sieber et al. 2016). Genes of this canonical axis ranked among the top NMD susceptible genes in all four primary cell types indicating that NMD constitutively restrains the integrated stress response as a basal function, extending findings discussed above made largely in transformed cell lines and rodent tissues to primary human lung cells.

RNA splicing was also a top-ranked enriched pathway in all four cell types. The leading-edge genes were predominantly SR proteins, hnRNPs, and core spliceosomal components. Nearly all human SR-protein genes, along with many other splicing regulators, produce unproductive mRNA isoforms through poison cassette exons or alternatively spliced 3′ UTR introns that are degraded by NMD, providing negative autoregulation of splicing-factor levels (Lareau et al. 2007; Ni et al. 2007). The exons that trigger this decay are frequently ultraconserved and the same regulation extends to core spliceosomal proteins such as SmB/B′ (*SNRPB*), which we likewise identify in leading edge analysis from our data (Saltzman et al. 2008). Consistent with this, 11 of the 12 canonical SR proteins (*SRSF1–SRSF11*) contained a significant NMD susceptible isoform in every cell type, and our pairwise analysis of isoform profiles from these genes identified the splicing events that would convert protein-coding isoforms to their NMD susceptible counterparts, demonstrating standard poison exon mechanisms for six of the SR family genes and alternative PTC-inducing mechanisms for the other five. More broadly, NMD susceptible genes were strongly enriched for RNA-binding proteins across all four cell types, by both the Gerstberger census (Gerstberger et al. 2014) and the orthogonal ENCODE eCLIP roster (Van Nostrand et al. 2020), with splicing regulators and core spliceosome components showing stronger enrichment than RBPs annotated as binding 3’

UTR regions. Because NMD itself limits the abundance of these regulators, inhibiting it stabilizes their unproductive isoforms which deregulates splice-factor levels and remodels the splicing program more broadly (Weischenfeldt et al. 2012). This is a recursive coupling, in which NMD both polices, and is shaped by, the splicing apparatus.

NMD is known to be context-sensitive, and there are many examples in which specific transcripts are degraded by NMD in a tissue-specific manner. However, most of these reports are focused on a single gene. Our analysis provides a transcriptome-wide view of NMD that allows us to estimate the degree of sharing versus cell-type specific NMD activity. We observed a substantial amount of sharing of NMD effects across cell types (44.8% at gene-level, 57.6% at isoform-level). On its surface, this differs from the previous report by Tan et al.[13] which reported only 16–25% transcript-level overlap of UPF2-dependent NMD targets between human embryonic stem cells and neural progenitor cells, but it is more in line with the study by Palou-Marquez et al[24]. However, re-analysis of the Tan data using comparable methods yielded a 41-50% transcript-level overlap that is more similar to our results. Regarding the cell type-specific aspects of NMD susceptibility, one possible explanation is that cell-type variation reflects differences in global NMD pathway activity rather than changes in substrate identity[24], though our study was not designed to address this.

Using a deep learning model, we could predict NMD susceptibility with high accuracy from isoform sequences and ORF-related annotations. Interpretable AI measures were used to interrogate these models to generate functional hypotheses, which verified known biology, gave additional information in areas of ongoing uncertainty, and suggested novel hypotheses. Our model clearly supported the role of PTCs as the most important mechanism of NMD susceptibility, and it also confirmed the role of 5’ UTR length and uORFs as a secondary mechanism of NMD involving ~10% of NMD susceptible isoforms. It shed light on a contested area, specifically the importance of 3’ UTR length as an NMD trigger, which did not appear to be a strong independent predictor of NMD in both our Isopair analysis and our predictive model, with the exception of instances where splicing within the 3’ UTR turned a normal stop codon into a PTC. The “faux 3’ UTR” postulated mechanism of NMD is supported by compelling data in yeast[1,25], but our data provides further support that the relationship of 3’UTRs to NMD in mammalian systems is more complex than the simple length-based relationship observed in yeast[26]. Finally, Shapley analysis of sequence features around stop codons suggests that NMD susceptible transcripts are also enriched for sequence elements that favor translational readthrough, such as increased use of the UGA stop codon as well as other supportive sequence features.

Our model differs from other NMD prediction models, NMDetective[19] and NMDEP[20], and compares favorably to them, likely because it is the only model trained on accurate isoform-level data provided by long reads. All three models explain a similar proportion of overall variance for NMD effects for isoforms present in our gene-matched isoform pairs analysis, reflecting the fact that the assignment of PTCs to the most highly expressed isoform in a gene was often correct. However, our model performs substantially better within isoform subgroups defined by NMD mechanisms, with markedly better performance in the NMD+/PTC- subgroup where 5’ UTR features appear to be driving NMD, reflecting the fact that our model learned a broader class of NMD-related mechanisms. The performance level of each of these models demonstrates that much can be learned from modeling NMD behavior from transcriptome-wide data, and future work in this area will likely improve our understanding of how to use interpretable machine learning methods to differentiate causal mechanisms from purely associative correlations.

The major strength of this paper is the use of long read RNA-seq coupled with targeted inhibition of NMD using a small molecule inhibitor of SMG1. This paper reveals for the first time tens of thousands of specific NMD targets across four cell types, which are publicly available at (https://nmd-lungcells.castaldilab.org/). Our data indicate that our experiments achieved effective inhibition of NMD with relatively few off-target effects. However, when we specifically analyzed the productive isoform subset within NMD and non-NMD targeted genes, there was some evidence of secondary responses to NMD inhibition that suggest that, at six hours, some downstream effects of NMD inhibition were evident in pathways heavily regulated by NMD. In the future, time series studies of multiple timepoints after NMD inhibition will allow for better delineation of direct and secondary effects of NMD inhibition. Analysis of four different primary lung cell types allowed for identification of a “core” set of NMD susceptible genes from more cell type-specific responses, and our findings of roughly 50% core versus 50% context-specific NMD fits into previous investigations of NMD across different cellular contexts. Our deep learning model confirms the ability to predict NMD susceptibility with good accuracy from sequence, and we demonstrate that Shapley analysis of these models can provide both high level and sequence specific insight into potential functional mechanisms underlying NMD, though our novel observations require further functional confirmation.

Cell Type	Number of Genes Tested	Number of Significant Genes (FDR<0.05)	NMD Susceptible Genes (FDR<0.05 & logFC>0)	Percent of Significant Genes that are NMD Susceptible	Mean logFC of NMD susceptible Genes
AT2	25,955	3,638	2,276	62.56%	0.80
LAE	25,955	6,753	3,326	49.25%	0.94
FB	25,955	3,122	2,066	66.18%	0.80
MV	25,955	3,428	2,310	67.39%	0.98

Table 1. Short-read Differential Gene Expression Results

Cell Type	Number of Transcripts Tested	Number of Significant Transcripts (FDR<0.05)	NMD Susceptible Transcripts (FDR<0.05 & logFC>0)	Percent of Significant Transcripts that are NMD Susceptible	Mean logFC of NMD Susceptible Transcripts

AT2	162,800	24,847	22,850	91.96%	2.09
LAE	162,800	35,336	32,239	91.24%	2.97
FB	162,800	24,803	22,552	90.92%	1.60
MV	162,800	27,834	25,294	90.87%	2.68

Table 2. Long-read Differential Isoform Expression Results

Cell type	Pathway	Database	NES	p.adj	Size
AT2	RNA splicing	GO:BP	1.49	0.018	462
	Vitamin D metabolic process	GO:BP	1.86	0.020	18
	tRNA modification	GO:BP	1.78	0.032	96
	RNA modification	GO:BP	1.59	0.045	178
	tRNA metabolic process	GO:BP	1.55	0.048	207
LAE	Cilium movement	GO:BP	1.69	0.002	249
	Cellular response to topologically incorrect protein	GO:BP	1.86	0.005	121
	Cilium- or flagellum-dependent cell motility	GO:BP	1.73	0.005	212
	Unfolded protein response	Reactome	1.92	0.007	92
	Response to topologically incorrect protein	GO:BP	1.75	0.007	168
FB	Cellular response to unfolded protein	GO:BP	1.71	0.040	106
	Cellular response to topologically incorrect protein	GO:BP	1.68	0.040	121
	Unfolded protein response	Reactome	1.73	0.044	92
MV	RNA splicing	GO:BP	1.57	<0.001	462
	Cellular response to topologically incorrect protein	GO:BP	1.88	0.001	121
	Response to topologically incorrect protein	GO:BP	1.78	0.002	168
	Cellular response to unfolded protein	GO:BP	1.88	0.002	106
	RNA splicing via transesterification	GO:BP	1.58	0.004	335

Table 3. Top 5 Gene Level Significant Pathways by Cell Type. FB only reached significance in 3 pathways.

Cell type	Pathway	Database	NES	p.adj	Size
AT2	RNA splicing	GO:BP	1.33	9.0×10^{-17}	438
	RNA splicing via transesterification reactions	GO:BP	1.34	7.9×10^{-14}	316
	Processing of capped intron-containing pre-mRNA	Reactome	1.33	1.6×10^{-11}	291
	Ribosome biogenesis	GO:BP	1.28	1.7×10^{-9}	325
	Translation	Reactome	1.26	3.3×10^{-9}	364
LAE	RNA splicing	GO:BP	1.33	1.2×10^{-17}	438
	Processing of capped intron-containing pre-mRNA	Reactome	1.39	4.5×10^{-17}	291
	Ribosome biogenesis	GO:BP	1.34	2.0×10^{-14}	325
	RNA splicing via transesterification reactions	GO:BP	1.35	2.0×10^{-14}	316
	Dengue virus infection	Reactome	1.32	1.6×10^{-13}	352
FB	RNA splicing	GO:BP	1.31	7.7×10^{-14}	438
	RNA splicing via transesterification reactions	GO:BP	1.31	5.2×10^{-11}	316
	Translation	Reactome	1.29	2.7×10^{-10}	364
	Ribosome biogenesis	GO:BP	1.30	7.0×10^{-10}	325
	Myc targets v1	Hallmark	1.35	4.7×10^{-9}	200
MV	RNA splicing	GO:BP	1.31	5.9×10^{-16}	438
	RNA splicing via transesterification reactions	GO:BP	1.32	3.1×10^{-13}	316
	Translation	Reactome	1.28	7.8×10^{-11}	364
	Processing of capped intron-containing pre-mRNA	Reactome	1.29	2.1×10^{-9}	291
	Ribosome biogenesis	GO:BP	1.26	1.4×10^{-8}	325

Table 4. Top 5 Isoform Level Significant Pathways by Cell Type.

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